

Annex A (normative): Measurement channels

A.1 General

The throughput values defined in the measurement channels specified in Annex A, are calculated and are valid per datastream (codeword). For multi-stream (more than one codeword) transmissions, the throughput referenced in the minimum requirements is the sum of throughputs of all datastreams (codewords).

The UE category entry in the definition of the reference measurement channel in Annex A is only informative and reveals the UE categories, which can support the corresponding measurement channel. Whether the measurement channel is used for testing a certain UE category or not is specified in the individual minimum requirements.

A.2 UL reference measurement channels

A.2.1 General

The measurement channels in the following clauses are defined to derive the requirements in clause 6 (Transmitter Characteristics) and clause 7 (Receiver Characteristics). The measurement channels represent example configurations of physical channels for different data rates. The measurement channels in the following clauses are applicable to both FDD and TDD.

The active uplink slots for TDD configurations are specified in table A.2.1-1. TDD slot patterns defined for reference sensitivity tests will be used for TDD UL RMCs.

Table A.2.1-1: TDD active uplink slots

SCS	Active Uplink slots
15 kHz	4, 9
30 kHz	8, 9, 18, 19
60 kHz	16, 17, 18, 19, 36, 37, 38, 39

A.2.2 Reference measurement channels

A.2.2.1 DFT-s-OFDM Pi/2-BPSK

Table A.2.2.1-1: Reference Channels for DFT-s-OFDM Pi/2-BPSK

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	pi/2 BPSK	0	24	16	2	1	132	132
	5	11	pi/2 BPSK	0	160	16	2	1	660	660
	9	11	pi/2 BPSK	0	288	16	2	1	1188	1188
	10	11	pi/2 BPSK	0	320	16	2	1	1320	1320
	12	11	pi/2 BPSK	0	384	16	2	1	1584	1584
	15	11	pi/2 BPSK	0	480	16	2	1	1980	1980
	18	11	pi/2 BPSK	0	576	16	2	1	2376	2376
	24	11	pi/2 BPSK	0	768	16	2	1	3168	3168
	25	11	pi/2 BPSK	0	808	16	2	1	3300	3300
	30	11	pi/2 BPSK	0	984	16	2	1	3960	3960
	32	11	pi/2 BPSK	0	1032	16	2	1	4224	4224
	36	11	pi/2 BPSK	0	1128	16	2	1	4752	4752
	45	11	pi/2 BPSK	0	1416	16	2	1	5940	5940
	50	11	pi/2 BPSK	0	1544	16	2	1	6600	6600
	60	11	pi/2 BPSK	0	1864	16	2	1	7920	7920
	64	11	pi/2 BPSK	0	2024	16	2	1	8448	8448
	75	11	pi/2 BPSK	0	2408	16	2	1	9900	9900
	80	11	pi/2 BPSK	0	2472	16	2	1	10560	10560
	81	11	pi/2 BPSK	0	2536	16	2	1	10692	10692
	90	11	pi/2 BPSK	0	2792	16	2	1	11880	11880
	100	11	pi/2 BPSK	0	3104	16	2	1	13200	13200
	108	11	pi/2 BPSK	0	3368	16	2	1	14256	14256
	120	11	pi/2 BPSK	0	3752	16	2	1	15840	15840
	128	11	pi/2 BPSK	0	3976	24	2	2	16896	16896

	135	11	pi/2 BPSK	0	4104	24	2	2	17820	17820
	160	11	pi/2 BPSK	0	4872	24	2	2	21120	21120
	162	11	pi/2 BPSK	0	5000	24	2	2	21384	21384
	180	11	pi/2 BPSK	0	5512	24	2	2	23760	23760
	216	11	pi/2 BPSK	0	6664	24	2	2	28512	28512
	243	11	pi/2 BPSK	0	7560	24	2	2	32076	32076
	270	11	pi/2 BPSK	0	8448	24	2	3	35640	35640
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted. NOTE 2: MCS Index is based on MCS table 6.1.4.1-1 defined in TS 38.214 [10]. NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.1-2: Void

Table A.2.2.1-3: Void

A.2.2.2 DFT-s-OFDM QPSK

Table A.2.2.2-1: Reference Channels for DFT-s-OFDM QPSK

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	QPSK	2	48	16	2	1	264	132
	5	11	QPSK	2	256	16	2	1	1320	660
	9	11	QPSK	2	456	16	2	1	2376	1188
	10	11	QPSK	2	504	16	2	1	2640	1320
	12	11	QPSK	2	608	16	2	1	3168	1584
	15	11	QPSK	2	768	16	2	1	3960	1980
	18	11	QPSK	2	928	16	2	1	4752	2376
	20	11	QPSK	2	1032	16	2	1	5280	2640
	24	11	QPSK	2	1192	16	2	1	6336	3168
	25	11	QPSK	2	1256	16	2	1	6600	3300
	30	11	QPSK	2	1544	16	2	1	7920	3960
	32	11	QPSK	2	1608	16	2	1	8448	4224
	36	11	QPSK	2	1800	16	2	1	9504	4752
	45	11	QPSK	2	2208	16	2	1	11880	5940
	50	11	QPSK	2	2472	16	2	1	13200	6600
	60	11	QPSK	2	3104	16	2	1	15840	7920
	64	11	QPSK	2	3240	16	2	1	16896	8448
	75	11	QPSK	2	3752	16	2	1	19800	9900
	80	11	QPSK	2	3976	24	2	2	21120	10560
	81	11	QPSK	2	4040	24	2	2	21384	10692
	90	11	QPSK	2	4488	24	2	2	23760	11880
	100	11	QPSK	2	5000	24	2	2	26400	13200
	108	11	QPSK	2	5384	24	2	2	28512	14256
	120	11	QPSK	2	5896	24	2	2	31680	15840
	128	11	QPSK	2	6408	24	2	2	33792	16896
	135	11	QPSK	2	6664	24	2	2	35640	17820
	160	11	QPSK	2	7944	24	2	3	42240	21120
	162	11	QPSK	2	8064	24	2	3	42768	21384
	180	11	QPSK	2	8976	24	2	3	47520	23760

	216	11	QPSK	2	10752	24	2	3	57024	28512
	243	11	QPSK	2	12040	24	2	4	64152	32076
	270	11	QPSK	2	13320	24	2	4	71280	35640
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted. NOTE 2: MCS Index is based on MCS table 6.1.4.1-1 defined in TS 38.214 [10]. NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.2-2: Void

Table A.2.2.2-3: Void

A.2.2.3 DFT-s-OFDM 16QAM

Table A.2.2.3-1: Reference Channels for DFT-s-OFDM 16QAM

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	16QAM	10	176	16	2	1	528	132
	5	11	16QAM	10	888	16	2	1	2640	660
	9	11	16QAM	10	1608	16	2	1	4752	1188
	10	11	16QAM	10	1800	16	2	1	5280	1320
	12	11	16QAM	10	2088	16	2	1	6336	1584
	15	11	16QAM	10	2664	16	2	1	7920	1980
	18	11	16QAM	10	3240	16	2	1	9504	2376
	24	11	16QAM	10	4224	24	1	1	12672	3168
	25	11	16QAM	10	4352	24	1	1	13200	3300
	30	11	16QAM	10	5248	24	1	1	15840	3960
	32	11	16QAM	10	5632	24	1	1	16896	4224
	36	11	16QAM	10	6272	24	1	1	19008	4752
	45	11	16QAM	10	7808	24	1	1	23760	5940
	50	11	16QAM	10	8712	24	1	2	26400	6600
	60	11	16QAM	10	10504	24	1	2	31680	7920
	64	11	16QAM	10	11272	24	1	2	33792	8448
	75	11	16QAM	10	13064	24	1	2	39600	9900
	80	11	16QAM	10	14088	24	1	2	42240	10560
	81	11	16QAM	10	14088	24	1	2	42768	10692
	90	11	16QAM	10	15880	24	1	2	47520	11880
	100	11	16QAM	10	17424	24	1	3	52800	13200
	108	11	16QAM	10	18960	24	1	3	57024	14256
	120	11	16QAM	10	21000	24	1	3	63360	15840
	128	11	16QAM	10	22536	24	1	3	67584	16896
	135	11	16QAM	10	23568	24	1	3	71280	17820
	160	11	16QAM	10	28168	24	1	4	84480	21120
	162	11	16QAM	10	28168	24	1	4	85536	21384
	180	11	16QAM	10	31752	24	1	4	95040	23760
	216	11	16QAM	10	37896	24	1	5	114048	28512

	243	11	16QAM	10	43032	24	1	6	128304	32076
	270	11	16QAM	10	47112	24	1	6	142560	35640
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted. NOTE 2: MCS Index is based on MCS table 6.1.4.1-1 defined in TS 38.214 [10]. NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.3-2: Void

Table A.2.2.3-3: Void

A.2.2.4 DFT-s-OFDM 64QAM

Table A.2.2.4-1: Reference Channels for DFT-s-OFDM 64QAM

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	64QAM	18	408	16	2	1	792	132
	5	11	64QAM	18	2024	16	2	1	3960	660
	9	11	64QAM	18	3624	16	2	1	7128	1188
	10	11	64QAM	18	3968	24	1	1	7920	1320
	12	11	64QAM	18	4736	24	1	1	9504	1584
	15	11	64QAM	18	6016	24	1	1	11880	1980
	18	11	64QAM	18	7168	24	1	1	14256	2376
	24	11	64QAM	18	9480	24	1	2	19008	3168
	25	11	64QAM	18	9992	24	1	2	19800	3300
	30	11	64QAM	18	12040	24	1	2	23760	3960
	32	11	64QAM	18	12808	24	1	2	25344	4224
	36	11	64QAM	18	14344	24	1	2	28512	4752
	45	11	64QAM	18	17928	24	1	3	35640	5940
	50	11	64QAM	18	19968	24	1	3	39600	6600
	60	11	64QAM	18	24072	24	1	3	47520	7920
	64	11	64QAM	18	25608	24	1	4	50688	8448
	75	11	64QAM	18	30216	24	1	4	59400	9900
	80	11	64QAM	18	31752	24	1	4	63360	10560
	81	11	64QAM	18	32264	24	1	4	64152	10692
	90	11	64QAM	18	35856	24	1	5	71280	11880
	100	11	64QAM	18	39936	24	1	5	79200	13200
	108	11	64QAM	18	43032	24	1	6	85536	14256
	120	11	64QAM	18	48168	24	1	6	95040	15840
	128	11	64QAM	18	51216	24	1	7	101376	16896
	135	11	64QAM	18	54296	24	1	7	106920	17820
	160	11	64QAM	18	63528	24	1	8	126720	21120
	162	11	64QAM	18	64552	24	1	8	128304	21384
	180	11	64QAM	18	71688	24	1	9	142560	23760
	216	11	64QAM	18	86040	24	1	11	171072	28512

	243	11	64QAM	18	96264	24	1	12	192456	32076
	270	11	64QAM	18	108552	24	1	13	213840	35640
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted. NOTE 2: MCS Index is based on MCS table 6.1.4.1-1 defined in TS 38.214 [10]. NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.4-2: Void

Table A.2.2.4-3: Void

A.2.2.5 DFT-s-OFDM 256QAM

Table A.2.2.5-1: Reference Channels for DFT-s-OFDM 256QAM

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	256QAM	20	704	16	2	1	1056	132
	5	11	256QAM	20	3496	16	2	1	5280	660
	9	11	256QAM	20	6272	24	1	1	9504	1188
	10	11	256QAM	20	7040	24	1	1	10560	1320
	12	11	256QAM	20	8456	24	1	2	12672	1584
	15	11	256QAM	20	10504	24	1	2	15840	1980
	18	11	256QAM	20	12552	24	1	2	19008	2376
	24	11	256QAM	20	16896	24	1	3	25344	3168
	25	11	256QAM	20	17424	24	1	3	26400	3300
	30	11	256QAM	20	21000	24	1	3	31680	3960
	32	11	256QAM	20	22536	24	1	3	33792	4224
	36	11	256QAM	20	25104	24	1	3	38016	4752
	45	11	256QAM	20	31752	24	1	4	47520	5940
	50	11	256QAM	20	34816	24	1	5	52800	6600
	60	11	256QAM	20	42016	24	1	5	63360	7920
	64	11	256QAM	20	45096	24	1	6	67584	8448
	75	11	256QAM	20	53288	24	1	7	79200	9900
	80	11	256QAM	20	56368	24	1	7	84480	10560
	81	11	256QAM	20	57376	24	1	7	85536	10692
	90	11	256QAM	20	63528	24	1	8	95040	11880
	100	11	256QAM	20	69672	24	1	9	105600	13200
	108	11	256QAM	20	75792	24	1	9	114048	14256
	120	11	256QAM	20	83976	24	1	10	126720	15840
	128	11	256QAM	20	90176	24	1	11	135168	16896
	135	11	256QAM	20	94248	24	1	12	142560	17820
	160	11	256QAM	20	112648	24	1	14	168960	21120
	162	11	256QAM	20	114776	24	1	14	171072	21384
	180	11	256QAM	20	127080	24	1	16	190080	23760
	216	11	256QAM	20	151608	24	1	18	228096	28512

	243	11	256QAM	20	172176	24	1	21	256608	32076
	270	11	256QAM	20	188576	24	1	23	285120	35640
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted. NOTE 2: MCS Index is based on MCS table 5.1.3.1-2 defined in TS 38.214 [10]. NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.5-2: Void

Table A.2.2.5-3: Void

A.2.2.6 CP-OFDM QPSK

Table A.2.2.6-1: Reference Channels for CP-OFDM QPSK

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	QPSK	2	48	16	2	1	264	132
	5	11	QPSK	2	256	16	2	1	1320	660
	6	11	QPSK	2	304	16	2	1	1584	792
	9	11	QPSK	2	456	16	2	1	2376	1188
	10	11	QPSK	2	504	16	2	1	2640	1320
	11	11	QPSK	2	552	16	2	1	2904	1452
	12	11	QPSK	2	608	16	2	1	3168	1584
	13	11	QPSK	2	672	16	2	1	3432	1716
	15	11	QPSK	2	768	16	2	1	3960	1980
	16	11	QPSK	2	808	16	2	1	4224	2112
	18	11	QPSK	2	928	16	2	1	4752	2376
	19	11	QPSK	2	984	16	2	1	5016	2508
	24	11	QPSK	2	1192	16	2	1	6336	3168
	25	11	QPSK	2	1256	16	2	1	6600	3300
	26	11	QPSK	2	1288	16	2	1	6864	3432
	31	11	QPSK	2	1544	16	2	1	8184	4092
	33	11	QPSK	2	1672	16	2	1	8712	4356
	38	11	QPSK	2	1928	16	2	1	10032	5016
	39	11	QPSK	2	2024	16	2	1	10296	5148
	40	11	QPSK	2	2024	16	2	1	10560	5280
	47	11	QPSK	2	2408	16	2	1	12408	6204
	51	11	QPSK	2	2536	16	2	1	13464	6732
	52	11	QPSK	2	2600	16	2	1	13728	6864
	53	11	QPSK	2	2664	16	2	1	13992	6996
	54	11	QPSK	2	2664	16	2	1	14256	7128
	61	11	QPSK	2	3104	16	2	1	16104	8052
	65	11	QPSK	2	3240	16	2	1	17160	8580
	67	11	QPSK	2	3368	16	2	1	17688	8844
	68	11	QPSK	2	3368	16	2	1	17952	8976

	78	11	QPSK	2	3848	24	2	2	20592	10296
	79	11	QPSK	2	3912	24	2	2	20856	10428
	80	11	QPSK	2	3976	24	2	2	21120	10560
	81	11	QPSK	2	4040	24	2	2	21384	10692
	93	11	QPSK	2	4616	24	2	2	24552	12276
	95	11	QPSK	2	4744	24	2	2	25080	12540
	106	11	QPSK	2	5256	24	2	2	27984	13992
	107	11	QPSK	2	5256	24	2	2	28248	14124
	108	11	QPSK	2	5384	24	2	2	28512	14256
	109	11	QPSK	2	5384	24	2	2	28776	14388
	121	11	QPSK	2	6024	24	2	2	31944	15972
	123	11	QPSK	2	6152	24	2	2	32472	16236
	133	11	QPSK	2	6664	24	2	2	35112	17556
	135	11	QPSK	2	6664	24	2	2	35640	17820
	137	11	QPSK	2	6792	24	2	2	36168	18084
	160	11	QPSK	2	7944	24	2	3	42240	21120
	162	11	QPSK	2	8064	24	2	3	42768	21384
	189	11	QPSK	2	9480	24	2	3	49896	24948
	216	11	QPSK	2	10752	24	2	3	57024	28512
	217	11	QPSK	2	10752	24	2	3	57288	28644
	245	11	QPSK	2	12296	24	2	4	64680	32340
	270	11	QPSK	2	13320	24	2	4	71280	35640
	273	11	QPSK	2	13576	24	2	4	72072	36036
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted. NOTE 2: MCS Index is based on MCS table 5.1.3.1-1 defined in TS 38.214 [10]. NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.6-2: Void

Table A.2.2.6-3: Void

A.2.2.7 CP-OFDM 16QAM

Table A.2.2.7-1: Reference Channels for CP-OFDM 16QAM

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	16QAM	10	176	16	2	1	528	132
	5	11	16QAM	10	888	16	2	1	2640	660
	6	11	16QAM	10	1064	16	2	1	3168	792
	9	11	16QAM	10	1608	16	2	1	4752	1188
	10	11	16QAM	10	1800	16	2	1	5280	1320
	11	11	16QAM	10	1928	16	2	1	5808	1452
	12	11	16QAM	10	2088	16	2	1	6336	1584
	13	11	16QAM	10	2280	16	2	1	6864	1716
	15	11	16QAM	10	2664	16	2	1	7920	1980
	16	11	16QAM	10	2792	16	2	1	8448	2112
	18	11	16QAM	10	3240	16	2	1	9504	2376
	19	11	16QAM	10	3368	16	2	1	10032	2508
	24	11	16QAM	10	4224	24	1	1	12672	3168
	25	11	16QAM	10	4352	24	1	1	13200	3300
	26	11	16QAM	10	4480	24	1	1	13728	3432
	31	11	16QAM	10	5376	24	1	1	16368	4092
	33	11	16QAM	10	5760	24	1	1	17424	4356
	38	11	16QAM	10	6656	24	1	1	20064	5016
	39	11	16QAM	10	6784	24	1	1	20592	5148
	40	11	16QAM	10	7040	24	1	1	21120	5280
	47	11	16QAM	10	8192	24	1	1	24816	6204
	51	11	16QAM	10	8968	24	1	2	26928	6732
	52	11	16QAM	10	9224	24	1	2	27456	6864
	53	11	16QAM	10	9224	24	1	2	27984	6996
	54	11	16QAM	10	9480	24	1	2	28512	7128
	61	11	16QAM	10	10760	24	1	2	32208	8052
	65	11	16QAM	10	11272	24	1	2	34320	8580
	67	11	16QAM	10	11784	24	1	2	35376	8844
	68	11	16QAM	10	11784	24	1	2	35904	8976

	78	11	16QAM	10	13576	24	1	2	41184	10296
	79	11	16QAM	10	13832	24	1	2	41712	10428
	80	11	16QAM	10	14088	24	1	2	42240	10560
	81	11	16QAM	10	14088	24	1	2	42768	10692
	93	11	16QAM	10	16392	24	1	2	49104	12276
	95	11	16QAM	10	16392	24	1	2	50160	12540
	106	11	16QAM	10	18432	24	1	3	55968	13992
	107	11	16QAM	10	18960	24	1	3	56496	14124
	108	11	16QAM	10	18960	24	1	3	57024	14256
	109	11	16QAM	10	18960	24	1	3	57552	14388
	121	11	16QAM	10	21000	24	1	3	63888	15972
	123	11	16QAM	10	21504	24	1	3	64944	16236
	133	11	16QAM	10	23040	24	1	3	70224	17556
	135	11	16QAM	10	23568	24	1	3	71280	17820
	137	11	16QAM	10	24072	24	1	3	72336	18084
	160	11	16QAM	10	28168	24	1	4	84480	21120
	162	11	16QAM	10	28168	24	1	4	85536	21384
	189	11	16QAM	10	32776	24	1	4	99792	24948
	216	11	16QAM	10	37896	24	1	5	114048	28512
	217	11	16QAM	10	37896	24	1	5	114576	28644
	245	11	16QAM	10	43032	24	1	6	129360	32340
	270	11	16QAM	10	47112	24	1	6	142560	35640
	273	11	16QAM	10	48168	24	1	6	144144	36036

NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted.

NOTE 2: MCS Index is based on MCS table 5.1.3.1-1 defined in TS 38.214 [10].

NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)

NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.

Table A.2.2.7-2: Void

Table A.2.2.7-3: Void

A.2.2.8 CP-OFDM 64QAM

Table A.2.2.8-1: Reference Channels for CP-OFDM 64QAM

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	64QAM	19	408	16	2	1	792	132
	5	11	64QAM	19	2024	16	2	1	3960	660
	9	11	64QAM	19	3624	16	2	1	7128	1188
	10	11	64QAM	19	3968	24	1	1	7920	1320
	11	11	64QAM	19	4352	24	1	1	8712	1452
	12	11	64QAM	19	4736	24	1	1	9504	1584
	13	11	64QAM	19	5120	24	1	1	10296	1716
	15	11	64QAM	19	6016	24	1	1	11880	1980
	18	11	64QAM	19	7168	24	1	1	14256	2376
	19	11	64QAM	19	7552	24	1		15048	2508
	24	11	64QAM	19	9480	24	1	2	19008	3168
	25	11	64QAM	19	9992	24	1	2	19800	3300
	26	11	64QAM	19	10504	24	1	2	20592	3432
	31	11	64QAM	19	12296	24	1	2	24552	4092
	33	11	64QAM	19	13064	24	1	2	26136	4356
	38	11	64QAM	19	15112	24	1	2	30096	5016
	39	11	64QAM	19	15624	24	1	2	30888	5148
	47	11	64QAM	19	18960	24	1	3	37224	6204
	51	11	64QAM	19	20496	24	1	3	40392	6732
	52	11	64QAM	19	21000	24	1	3	41184	6864
	53	11	64QAM	19	21000	24	1	3	41976	6996
	61	11	64QAM	19	24567	24	1	3	48312	8052
	65	11	64QAM	19	26120	24	1	4	51480	8580
	67	11	64QAM	19	26632	24	1	4	53064	8844
	78	11	64QAM	19	31240	24	1	4	61776	10296
	79	11	64QAM	19	31752	24	1	4	62568	10428
	80	11	64QAM	19	31752	24	1	4	63360	10560
	81	11	64QAM	19	32264	24	1	4	64152	10692
	93	11	64QAM	19	36896	24	1	5	73656	12276

	95	11	64QAM	19	37896	24	1	5	75240	12540
	106	11	64QAM	19	42016	24	1	5	83952	13992
	107	11	64QAM	19	43032	24	1	6	84744	14124
	108	11	64QAM	19	43032	24	1	6	85536	14256
	109	11	64QAM	19	44040	24	1	6	86328	14388
	121	11	64QAM	19	48168	24	1	6	95832	15972
	123	11	64QAM	19	49176	24	1	6	97416	16236
	133	11	64QAM	19	53288	24	1	7	105336	17556
	135	11	64QAM	19	54296	24	1	7	106920	17820
	137	11	64QAM	19	54296	24	1	7	108504	18084
	160	11	64QAM	19	63528	24	1	8	126720	21120
	162	11	64QAM	19	64552	24	1	8	128304	21384
	189	11	64QAM	19	75792	24	1	9	149688	24948
	216	11	64QAM	19	86040	24	1	11	171072	28512
	217	11	64QAM	19	86040	24	1	11	171864	28644
	245	11	64QAM	19	98376	24	1	12	194040	32340
	270	11	64QAM	19	108552	24	1	13	213840	35640
	273	11	64QAM	19	108552	24	1	13	216216	36036
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted. NOTE 2: MCS Index is based on MCS table 5.1.3.1-1 defined in TS 38.214 [10]. NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit) NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.8-2: Void

Table A.2.2.8-3: Void

A.2.2.9 CP-OFDM 256QAM

Table A.2.2.9-1: Reference Channels for CP-OFDM 256QAM

Parameter	Allocated resource blocks (L_{CRB})	DFT-s-OFDM Symbols per slot (Note 1)	Modulation	MCS Index (Note 2)	Payload size	Transport block CRC	LDPC Base Graph	Number of code blocks per slot (Note 3)	Total number of bits per slot	Total modulated symbols per slot
Unit					Bits	Bits			Bits	
	1	11	256QAM	20	704	16	2	1	1056	132
	5	11	256QAM	20	3496	16	2	1	5280	660
	9	11	256QAM	20	6272	24	1	1	9504	1188
	10	11	256QAM	20	7040	24	1	1	10560	1320
	11	11	256QAM	20	7680	24	1	1	11616	1452
	12	11	256QAM	20	8456	24	1	2	12672	1584
	13	11	256QAM	20	9224	24	1	2	13728	1716
	15	11	256QAM	20	10504	24	1	2	15840	1980
	18	11	256QAM	20	12552	24	1	2	19008	2376
	19	11	256QAM	20	13320	24	1	2	20064	2508
	24	11	256QAM	20	16896	24	1	3	25344	3168
	25	11	256QAM	20	17424	24	1	3	26400	3300
	26	11	256QAM	20	18432	24	1	3	27456	3432
	31	11	256QAM	20	22032	24	1	3	32736	4092
	33	11	256QAM	20	23040	24	1	3	34848	4356
	38	11	256QAM	20	26632	24	1	4	40128	5016
	39	11	256QAM	20	27656	24	1	4	41184	5148
	47	11	256QAM	20	32776	24	1	4	49632	6204
	51	11	256QAM	20	35856	24	1	5	53856	6732
	52	11	256QAM	20	36896	24	1	5	54912	6864
	53	11	256QAM	20	36896	24	1	5	55968	6996
	61	11	256QAM	20	43032	24	1	6	64416	8052
	65	11	256QAM	20	46104	24	1	6	68640	8580
	67	11	256QAM	20	47112	24	1	6	70752	8844
	78	11	256QAM	20	55304	24	1	7	82368	10296
	79	11	256QAM	20	55304	24	1	7	83424	10428
	80	11	256QAM	20	56368	24	1	7	84480	10560
	81	11	256QAM	20	57376	24	1	7	85536	10692

	93	11	256QAM	20	65576	24	1	8	98208	12276
	95	11	256QAM	20	67584	24	1	8	100320	12540
	106	11	256QAM	20	73776	24	1	9	111936	13992
	107	11	256QAM	20	75792	24	1	9	112992	14124
	108	11	256QAM	20	75792	24	1	9	114048	14256
	109	11	256QAM	20	75792	24	1	9	115104	14388
	121	11	256QAM	20	86040	24	1	11	127776	15972
	123	11	256QAM	20	86040	24	1	11	129888	16236
	133	11	256QAM	20	94248	24	1	12	140448	17556
	135	11	256QAM	20	94248	24	1	12	142560	17820
	137	11	256QAM	20	96264	24	1	12	144672	18084
	160	11	256QAM	20	112648	24	1	14	168960	21120
	162	11	256QAM	20	114776	24	1	14	171072	21384
	189	11	256QAM	20	131176	24	1	16	199584	24948
	216	11	256QAM	20	151608	24	1	18	228096	28512
	217	11	256QAM	20	151608	24	1	18	229152	28644
	245	11	256QAM	20	172176	24	1	21	258720	32340
	270	11	256QAM	20	188576	24	1	23	285120	35640
	273	11	256QAM	20	192624	24	1	23	288288	36036
NOTE 1: PUSCH mapping Type-A and single-symbol DM-RS configuration Type-1 with 2 additional DM-RS symbols, such that the DM-RS positions are set to symbols 2, 7, 11. DMRS is [TDM'ed] with PUSCH data. DM-RS symbols are not counted.										
NOTE 2: MCS Index is based on MCS table 5.1.3.1-2 defined in TS 38.214 [10].										
NOTE 3: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit)										
NOTE 4: The RMCs apply to all channel bandwidth where $L_{CRB} \leq N_{RB}$.										

Table A.2.2.9-2: Void

Table A.2.2.9-3: Void

A.2.3 Reference measurement channels for TDD

The TDD UL RMCs are defined in clause A.2.2 with the active UL slots specified in table A.2.1-1 and TDD slot patterns as defined for reference sensitivity tests.

A.2.3.1 DFT-s-OFDM Pi/2-BPSK

Table A.2.3.1-1: Void

Table A.2.3.1-2: Void

Table A.2.3.1-3: Void

A.2.3.2 DFT-s-OFDM QPSK

Table A.2.3.2-1: Void

Table A.2.3.2-2: Void

Table A.2.3.2-3: Void

A.2.3.3 DFT-s-OFDM 16QAM

Table A.2.3.3-1: Void

Table A.2.3.3-2: Void

Table A.2.3.3-3: Void

A.2.3.4 DFT-s-OFDM 64QAM

Table A.2.3.4-1: Void

Table A.2.3.4-2: Void

Table A.2.3.4-3: Void

A.2.3.5 DFT-s-OFDM 256QAM

Table A.2.3.5-1: Void

Table A.2.3.5-2: Void

Table A.2.3.5-3: Void

A.2.3.6 CP-OFDM QPSK

Table A.2.3.6-1: Void

Table A.2.3.6-2: Void

Table A.2.3.6-3: Void

A.2.3.7 CP-OFDM 16QAM

Table A.2.3.7-1: Void

Table A.2.3.7-2: Void

Table A.2.3.7-3: Void

A.2.3.8 CP-OFDM 64QAM

Table A.2.3.8-1: Void

Table A.2.3.8-2: Void

Table A.2.3.8-3: Void

A.2.3.9 CP-OFDM 256QAM

Table A.2.3.9-1: Void

Table A.2.3.9-2: Void

Table A.2.3.9-3: Void

A.3 DL reference measurement channels

A.3.1 General

Unless otherwise stated, Tables A.3.2.2-1, A.3.2.2-2, A.3.2.2-3, A.3.3.2-1, A.3.3.2-2 and A.3.3.2-3 are applicable for measurements of the Receiver Characteristics (clause 7) with the exception of clauses 7.4 (Maximum input level).

Unless otherwise stated, Tables A.3.2.3-1, A.3.2.3-2, A.3.2.3-3, A.3.3.3-1, A.3.3.3-2 and A.3.3.3-3 are applicable for clauses 7.4 (Maximum input level) and for UE not supporting PDSCH 256QAM,

Unless otherwise stated, Tables A.3.2.4-1, A.3.2.4-2, A.3.2.4-3, A.3.3.4-1, A.3.3.4-2 and A.3.3.4-3 are applicable for clauses 7.4 (Maximum input level) and for UE supporting PDSCH 256QAM,

Unless otherwise stated, Tables A.3.2.2-1, A.3.2.2-2, A.3.2.2-3, A.3.3.2-1, A.3.3.2-2 and A.3.3.2-3 also apply for the modulated interferer used in Clauses 7.5, 7.6 and 7.8 with test specific bandwidths.

In case of carrier aggregation scenarios, the k1 values and number of HARQ processes of the Reference Measurement Channels specified in Annex A.3 shall be adapted as specified in table A.3.1-2 and A.3.1-3.

Table A.3.1-1: Common reference channel parameters

Parameter		Unit	Value
CORESET frequency domain allocation			Full BW
CORESET time domain allocation			2 OFDM symbols at the begin of each slot
PDSCH mapping type			Type A
PDSCH start symbol index (S)			2
Number of consecutive PDSCH symbols (L)			12
PDSCH PRB bundling		PRBs	2
Dynamic PRB bundling			false
Overhead value for TBS determination			0
First DMRS position for Type A PDSCH mapping			2
DMRS type			Type 1
Number of additional DMRS			2
FDM between DMRS and PDSCH			Disable
CSIRS for tracking	First subcarrier index in the PRB used for CSI-RS (k0)		0 for CSI-RS resource 1,2,3,4
	OFDM symbols in the PRB used for CSIRS		$l_0 = 6$ for CSI-RS resource 1 and 3 $l_0 = 10$ for CSI-RS resource 2 and 4

	Number of CSI-RS ports		1 for CSI-RS resource 1,2,3,4
	CDM Type		'No CDM' for CSI-RS resource 1,2,3,4
	Density (p)		3 for CSI-RS resource 1,2,3,4
	CSIRS periodicity	Slots	15 kHz SCS: 20 for CSI-RS resource 1,2,3,4 30 kHz SCS: 40 for CSI-RS resource 1,2,3,4 60 kHz SCS: 80 for CSI-RS resource 1,2,3,4
	CSIRS offset	Slots	15 kHz SCS: 0 for CSI-RS resource 1 and 2 1 for CSI-RS resource 3 and 4 30 kHz SCS: 1 for CSI-RS resource 1 and 2 2 for CSI-RS resource 3 and 4 60 kHz SCS: 2 for CSI-RS resource 1 and 2 3 for CSI-RS resource 3 and 4
	Frequency Occupation		Start PRB 0 Number of PRB = BWP size
	QCL info		TCI state #0
PTRS configuration			PTRS is not configured

Table A.3.1-2: Carrier aggregation test parameters for K1 values

The number of slots between PDSCH and corresponding HARQ-ACK information		CCs with the same duplex mode and SCS with Pcell	CCs with different duplex mode and/or SCS with Pcell
FDD 15 kHz +	FDD PCell	{2}	N/A
FDD 15 kHz CA			
FDD 15 kHz +	15kHz PCell	{2}	{3}
FDD 30 kHz CA	30kHz PCell	{2}	{2}
FDD 15 kHz +	FDD PCell	{2}	{2}
TDD 15 kHz CA	TDD PCell	{4,3,2}	{4,3,2,6,5}
FDD 15 kHz +	FDD PCell	{2}	{3}
TDD 30 kHz CA	TDD PCell	{8,7,6,5,4,3,2}	{8,6,4,2,10}
TDD 15 kHz +	TDD PCell	{4,3,2}	N/A

TDD 15 kHz CA			
TDD 15 kHz +	15kHz PCell	{4,3,2}	{4,4,3,3,2,7,6}
TDD 30 kHz CA	30kHz PCell	{8,7,6,5,4,3,2}	{7,5,4}
FDD 30 kHz +	FDD PCell	{2}	N/A
FDD 30 kHz CA			
FDD 30 kHz +	FDD PCell	{2}	{2}
TDD 15 kHz CA	TDD PCell	{4,3,2}	{4,4,3,3,7,7,6,6,5,5}
FDD 30 kHz +	FDD PCell	{2}	{2}
TDD 30 kHz CA	TDD PCell	{8,7,6,5,4,3,2}	{8,7,6,5,4,3,2,2,10,-}(NOTE 1)
TDD 30 kHz +	TDD PCell	{8,7,6,5,4,3,2}	N/A
TDD 30 kHz CA			
NOTE 1: No PDSCH shall be scheduled in slots 9 and 19 to avoid HARQ conflicts and maximize Throughput. Hence no K1 value is applicable for them.			

Table A.3.1-3: Carrier Aggregation test parameters for number of HARQ processes

HARQ process number		CCs with the same duplex mode and SCS with Pcell	CCs with different duplex mode and/or SCS with Pcell
FDD 15 kHz +	FDD PCell	4	N/A
FDD 15 kHz CA			
FDD 15 kHz +	15kHz PCell	8	8
FDD 30 kHz CA	30kHz PCell	8	8
FDD 15 kHz +	FDD PCell	4	8
TDD 15 kHz CA	TDD PCell	8	8
FDD 15 kHz +	FDD PCell	4	8
TDD 30 kHz CA	TDD PCell	10	8
TDD 15 kHz +	TDD PCell	8	N/A
TDD 15 kHz CA			
TDD 15 kHz +	15kHz PCell	8	12
TDD 30 kHz CA	30kHz PCell	8	8
FDD 30 kHz +	FDD PCell	8	N/A

FDD 30 kHz CA			
FDD 30 kHz +	FDD PCell	8	8
TDD 15 kHz CA	TDD PCell	8	16
FDD 30 kHz +	FDD PCell	8	8
TDD 30 kHz CA	TDD PCell	8	16
TDD 30 kHz +	TDD PCell	8	N/A
TDD 30 kHz CA			

A.3.2 DL reference measurement channels for FDD

A.3.2.1 General

Table A.3.2.1-1 Additional reference channels parameters for FDD

Parameter	Unit	Value
Number of HARQ Processes		4
K1 value		2 for all slots

A.3.2.2 FRC for receiver requirements for QPSK

Table A.3.2.2-1 Fixed reference channel for receiver requirements (SCS 15 kHz, FDD, QPSK 1/3)

Parameter	Unit	Value							
Channel bandwidth	MHz	5	10	15	20	25	30	40	50
Subcarrier spacing	kHz	15	15	15	15	15	15	15	15
Subcarrier spacing configuration		0	0	0	0	0	0	0	0
Allocated resource blocks		25	52	79	106	133	160	216	270
Subcarriers per resource block		12	12	12	12	12	12	12	12
Allocated slots per Frame		8	8	8	8	8	8	8	8
MCS Index		4	4	4	4	4	4	4	4
MCS Table for TBS determination		64QAM							
Modulation		QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK
Target Coding Rate		1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1

Information Bit Payload per Slot									
For Slots 0,1	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	Bits	1672	3368	5120	6912	8712	10504	14088	17424
Transport block CRC	Bits	16	16	24	24	24	24	24	24
LDPC base graph		2	2	1	1	1	1	1	1
Number of Code Blocks per Slot									
For Slots 0,1	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	CBs	1	1	1	1	2	2	2	3
Binary Channel Bits per Slot									
For Slots 0,1	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	Bits	5400	11232	17064	22896	28728	34560	46656	58320
Max. Throughput averaged over 1 frame	Mbps	1.338	2.694	4.096	5.530	6.970	8.403	11.270	13.9392
NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.									
NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).									
NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame									
NOTE 4: Slot i is slot index per frame									

Table A.3.2.2-2 Fixed reference channel for receiver requirements (SCS 30 kHz, FDD, QPSK 1/3)

Parameter	Unit	Value										
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	90	100
Subcarrier spacing configuration		1	1	1	1	1	1	1	1	1	1	1
Allocated resource blocks		24	38	51	65	78	106	133	162	217	245	273
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		17	17	17	17	17	17	17	17	17	17	17
MCS Index		4	4	4	4	4	4	4	4	4	4	4
MCS Table for TBS determination		64QAM										
Modulation		QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK
Target Coding Rate		1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot												
For Slots 0,1,2	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 3,...,19	Bits	1608	2472	3368	4224	4992	6912	8712	10504	14088	15880	17928
Transport block CRC	Bits	16	16	16	24	24	24	24	24	24	24	24
LDPC base graph		2	2	2	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot												
For Slots 0,1,2	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 3,...,19	CBs	1	1	1	1	1	1	2	2	2	2	3
Binary Channel Bits per Slot												
For Slot 0,1,2	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 3,...,19	Bits	5184	8208	11016	14040	16848	22896	28728	34992	46872	52920	58968
Max. Throughput averaged over 1 frame	Mbps	2.734	4.202	5.726	7.181	8.486	11.750	14.810	17.857	23.950	26.996	30.478

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.

NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).

NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame

NOTE 4: Slot i is slot index per frame

Table A.3.2.2-3 Fixed reference channel for receiver requirements (SCS 60 kHz, FDD, QPSK 1/3)

Parameter	Unit	Value										
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	90	100
Subcarrier spacing configuration		2	2	2	2	2	2	2	2	2	2	2
Allocated resource blocks		11	18	24	31	38	51	65	79	107	121	135
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		36	36	36	36	36	36	36	36	36	36	36
MCS Index		4	4	4	4	4	4	4	4	4	4	4
MCS Table for TBS Determination		64QAM										
Modulation		QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK
Target Coding Rate		1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot												
For Slots 0,1,2,3	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	Bits	736	1192	1608	2024	2472	3368	4224	5120	6912	7808	8712
Transport block CRC	Bits	16	16	16	16	16	16	24	24	24	24	24
LDPC base graph		2	2	2	2	2	2	1	1	1	1	1
Number of Code Blocks per Slot												
For Slots 0,1,2,3	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	CBs	1	1	1	1	1	1	1	1	1	1	2
Binary Channel Bits per Slot												
For Slots 0,1,2,3	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	Bits	2376	3888	5184	6696	8208	11016	14040	17064	23112	26136	29160
Max. Throughput averaged over 1 frame	Mbps	2.650	4.291	5.789	7.286	8.899	12.125	15.206	18.432	24.883	28.109	31.363

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.

NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).

NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame

NOTE 4: Slot i is slot index per frame

A.3.2.3 FRC for maximum input level for 64QAM

Table A.3.2.3-1 Fixed reference channel for maximum input level receiver requirements (SCS 15 kHz, FDD, 64QAM)

Parameter	Unit	Value							
Channel bandwidth	MHz	5	10	15	20	25	30	40	50
Subcarrier spacing	kHz	15	15	15	15	15	15	15	15
Subcarrier spacing configuration		0	0	0	0	0	0	0	0
Allocated resource blocks		25	52	79	106	133	[160]	216	270
Subcarriers per resource block		12	12	12	12	12	12	12	12
Allocated slots per Frame		8	8	8	8	8	8	8	8
MCS Index		24	24	24	24	24	24	24	24
MCS Table for TBS determination		64QAM							
Modulation		64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM
Target Coding Rate		3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1
Information Bit Payload per Slot									
For Slots 0,1	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	Bits	12296	25608	38936	52224	64552	77896	106576	131176
Transport block CRC	Bits	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1
Number of Code Blocks per Slot									
For Slots 0,1	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	CBs	2	4	5	7	8	10	13	16
Binary Channel Bits per Slot									
For Slots 0,1	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	Bits	16200	33696	51192	68688	86184	103680	139968	174960
Max. Throughput averaged over 1 frame	Mbps	9.837	20.486	31.149	41.779	51.642	62.317	85.261	104.941

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.

NOTE 2: If more than one Code Block is present, an additional CRC sequence of $L = 24$ Bits is attached to each Code Block (otherwise $L = 0$ Bit).

NOTE 3: SS/PBCH block is transmitted in slot 0 of each frame

NOTE 4: Slot i is slot index per frame

Table A.3.2.3-3 Fixed Reference Channel for Maximum input level receiver requirements (SCS 60 kHz, FDD, 64QAM)

Parameter	Unit	Value									
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	100
Subcarrier spacing configuration		2	2	2	2	2	2	2	2	2	2
Allocated resource blocks		11	18	24	31	38	51	65	79	107	135
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		36	36	36	36	36	36	36	36	36	36
MCS Index		24	24	24	24	24	24	24	24	24	24
MCS Table for TBS determination		64QAM									
Modulation		64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM
Target Coding Rate		3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot											
For Slots 0,1,2,3	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	Bits	5376	8712	11784	15112	18432	25104	31752	38936	52224	65576
Transport block CRC	Bits	24	24	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot											
For Slots 0,1,2,3	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	CBs	1	2	2	2	3	3	4	5	7	8
Binary Channel Bits per Slot											
For Slots 0,1,2,3	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	Bits	7128	11664	15552	20088	24624	33048	42120	51192	69336	87480
Max. Throughput averaged over 1 frame	Mbps	19.354	31.363	42.422	54.403	66.355	90.374	114.307	140.170	188.006	236.074

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.
NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame
NOTE 4: Slot i is slot index per frame

A.3.2.4 FRC for maximum input level for 256 QAM

Table A.3.2.4-1 Fixed reference channel for maximum input level receiver requirements (SCS 15 kHz, FDD, 256QAM)

Parameter	Unit	Value							
Channel bandwidth	MHz	5	10	15	20	25	30	40	50
Subcarrier spacing	kHz	15	15	15	15	15	15	15	15
Subcarrier spacing configuration		0	0	0	0	0	0	0	0
Allocated resource blocks		25	52	79	106	133	160	216	270
Subcarriers per resource block		12	12	12	12	12	12	12	12
Allocated slots per Frame		8	8	8	8	8	8	8	8
MCS Index		23	23	23	23	23	23	23	23
MCS Table for TBS determination		256QAM							
Modulation		256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM
Target Coding Rate		4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1
Information Bit Payload per Slot									
For Slots 0,1	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	Bits	16896	34816	53288	71688	90176	108552	143400	180376
Transport block CRC	Bits	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1
Number of Code Blocks per Slot									
For Slots 0,1	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	CBs	3	5	7	9	11	13	18	22
Binary Channel Bits per Slot									
For Slots 0,1	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,3,4,5,6,7,8,9	Bits	21600	44928	68256	91584	114912	138240	186624	233280
Max. Throughput averaged over 1 frame	Mbps	13.517	27.853	42.630	57.350	72.141	86.842	114.720	144.301
NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.									
NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).									
NOTE 3: SS/PBCH block is transmitted in slot 0 of each frame									
NOTE 4: Slot i is slot index per frame									

Table A.3.2.4-2 Fixed reference channel for maximum input level receiver requirements (SCS 30 kHz, FDD, 256QAM)

Parameter	Unit	Value									
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	100
Subcarrier spacing configuration		1	1	1	1	1	1	1	1	1	1
Allocated resource blocks		24	38	51	65	78	106	133	162	217	273
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		17	17	17	17	17	17	17	17	17	17
MCS Index		23	23	23	23	23	23	23	23	23	23
MCS Table for TBS determination		256QAM									
Modulation		256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM
Target Coding Rate		4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot											
For Slots 0,1,2	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 3,...,19	Bits	16136	25608	33816	44040	52224	71688	90176	108552	147576	184424
Transport block CRC	Bits	24	24	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot											
For Slots 0,1,2	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 3,...,19	CBs	2	4	5	6	7	9	11	13	18	22
Binary Channel Bits per Slot											
For Slots 0,1,2	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 3,...,19	Bits	20736	32832	44064	56160	67392	91584	114912	139968	187488	235872
Max. Throughput averaged over 1 frame	Mbps	27.431	43.534	57.487	74.868	88.781	121.870	153.299	184.538	250.879	313.521

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.

NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).

NOTE 3: SS/PBCH block is transmitted in slot 0 of each frame

NOTE 4: Slot i is slot index per frame

Table A.3.2.4-3 Fixed reference channel for maximum input level receiver requirements (SCS 60 kHz, FDD, 256QAM)

[illegible]

Allocated slots per Frame		36	36	36	36	36	36	36	36	36	36
MCS Index		23	23	23	23	23	23	23	23	23	23
MCS Table for TBS determination		256QAM									
Modulation		256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM
Target Coding Rate		4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot											
For Slots 0,1,2,3	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	Bits	7424	12040	16136	21000	25608	33816	44040	53288	71688	90176
Transport block CRC	Bits	24	24	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot											
For Slots 0,1,2,3	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	CBs	1	2	2	3	4	5	6	7	9	11
Binary Channel Bits per Slot											
For Slots 0,1,2,3	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 4,...,39	Bits	9504	15552	20736	26784	32832	44064	56160	68256	92448	116640
Max. Throughput averaged over 1 frame	Mbps	26.726	43.344	58.090	75.600	92.189	121.738	158.544	191.837	258.077	324.634
NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.2.1-1.											
NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).											
NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame											
NOTE 4: Slot i is slot index per frame											

A.3.3 DL reference measurement channels for TDD

A.3.3.1 General

Table A.3.3.1-1 Additional reference channels parameters for TDD

Parameter		Value		
		SCS 15 kHz ($\mu=0$)	SCS 30 kHz ($\mu=1$)	SCS 60 kHz ($\mu=2$)
TDD Slot Configuration pattern (Note 1)		DDDSU	7DS2U	14DS ₁ S ₂ 4U
Special Slot Configuration (Note 2)		10D+2G+2U	6D+4G+4U	S ₁ =12D+2G, S ₂ =6G+8U
referenceSubcarrierSpacing		15 kHz	30 kHz	60 kHz
UL-DL configuration	<i>dl-UL-TransmissionPeriodicity</i>	5 ms	5 ms	5 ms
	<i>nrofDownlinkSlots</i>	3	7	14
	<i>nrofDownlinkSymbols</i>	10	6	12
	<i>nrofUplinkSlot</i>	1	2	4
	<i>nrofUplinkSymbols</i>	2	4	8
Number of HARQ Processes		8	8	16
The number of slots between PDSCH and corresponding HARQ-ACK information (Note 3)		K1 = 4 if mod(i,5) = 0 K1 = 3 if mod(i,5) = 1 K1 = 2 if mod(i,5) = 2 where i is slot index per frame; i = {0,...,9}	K1 = 8 if mod(i,10) = 0 K1 = 7 if mod(i,10) = 1 K1 = 6 if mod(i,10) = 2 K1 = 5 if mod(i,10) = 3 K1 = 4 if mod(i,10) = 4 K1 = 3 if mod(i,10) = 5 K1 = 2 if mod(i,10) = 6 where i is slot index per frame; i = {0,...,19}	K1 = 13 if mod(i,20) = 2 K1 = 12 if mod(i,20) = 3 K1 = 11 if mod(i,20) = 4 K1 = 10 if mod(i,20) = 5 K1 = 9 if mod(i,20) = 6 K1 = 8 if mod(i,20) = 7 K1 = 7 if mod(i,20) = 8 K1 = 6 if mod(i,20) = 9 K1 = 6 if mod(i,20) = 10 K1 = 6 if mod(i,20) = 11 K1 = 6 if mod(i,20) = 12 K1 = 6 if mod(i,20) = 13 where i is slot index per frame; i = {0,...,39}
NOTE 1: D denotes a slot with all DL symbols; S denotes a slot with a mix of DL, UL and guard symbols; U denotes a slot with all UL symbols. The field is for information. NOTE 2: D, G, U denote DL, guard and UL symbols, respectively. The field is for information. NOTE 3: i is the slot index per frame. NOTE 4: A -2ms or +3ms time offset to the NR configuration pattern relative to the E-UTRA UL-DL configuration must be apply				

in the TDD intra-band EN-DC.

A.3.3.2 FRC for receiver requirements for QPSK

Table A.3.3.2-1 Fixed reference channel for receiver requirements (SCS 15 kHz, TDD, QPSK 1/3)

Parameter	Unit	Value							
Channel bandwidth	MHz	5	10	15	20	25	30	40	50
Subcarrier spacing	kHz	15	15	15	15	15	15	15	15
Subcarrier spacing configuration μ		0	0	0	0	0	0	0	0
Allocated resource blocks		25	52	79	106	133	160	216	270
Subcarriers per resource block		12	12	12	12	12	12	12	12
Allocated slots per Frame		4	4	4	4	4	4	4	4
MCS Index		4	4	4	4	4	4	4	4
MCS Table for TBS determination		64QAM							
Modulation		QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK
Target Coding Rate		1/3	1/3	1/3	1/3	1/3	1/3	1/3	1/3
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1
Information Bit Payload per Slot									
For Slots 0,1,3,4,8,9	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,5,6,7	Bits	1672	3368	5120	6912	8712	10504	14088	17424
Transport block CRC	Bits	16	16	24	24	24	24	24	24
LDPC base graph		2	2	1	1	1	1	1	1
Number of Code Blocks per Slot									
For Slots 0,1,3,4,8,9	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,5,6,7	CBs	1	1	1	1	2	2	2	3
Binary Channel Bits per Slot									
For Slots 0,1,3,4,8,9	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slots 2,5,6,7	Bits	5400	11232	17064	22896	28728	34560	46656	58320

Max. Throughput averaged over 1 frame	Mbps	0.669	1.347	2.048	2.765	3.485	4.202	5.635	6.970
NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.3.1-1.									
NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).									
NOTE 3: SS/PBCH block is transmitted in slot 0 of each frame									
NOTE 4: Slot i is slot index per frame									

Table A.3.3.2-3 Fixed reference channel for receiver requirements (SCS 60 kHz, TDD, QPSK 1/3)

[illegible]

A.3.3.3 FRC for maximum input level for 64QAM

Table A.3.3.3-1 Fixed reference channel for maximum input level receiver requirements (SCS 15 kHz, TDD, 64QAM)

[illegible]

Table A.3.3.3-2 Fixed reference channel for maximum input level receiver requirements (SCS 30 kHz, TDD, 64QAM)

Parameter	Unit	Value									
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	100
Subcarrier spacing configuration		1	1	1	1	1	1	1	1	1	1
Allocated resource blocks		24	38	51	65	78	106	133	162	217	273
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		11	11	11	11	11	11	11	11	11	11
MCS Index		24	24	24	24	24	24	24	24	24	24
MCS Table for TBS determination		64QAM									
Modulation		64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM
Target Coding Rate		3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot											
For Slots 0,1,2 and Slot i, if mod(i, 10) = {7,8,9} for i from {0,...,19}	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {13,...,19}	Bits	11784	18432	25104	31752	37896	52224	64552	79896	106576	135296
Transport block CRC	Bits	24	24	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot											
For Slots 0,1,2 and Slot i, if mod(i, 10) = {7,8,9} for i from {0,...,19}	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {3,...,19}	CBs	2	3	3	4	5	7	8	10	13	17
Binary Channel Bits per Slot											
For Slots 0,1,2 and Slot i, if mod(i, 10) = {7,8,9} for i from {0,...,19}	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 10) = {0,1,2,3,4,5,6} for i from {3,...,19}	Bits	15552	24624	33048	42120	50544	68688	86184	104976	140616	176904
Max. Throughput averaged over 1 frame	Mbps	12.962	20.275	27.614	34.927	41.686	57.446	71.007	87.886	117.234	148.826

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.3.1-1.
NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).
NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame
NOTE 4: Slot i is slot index per frame

Table A.3.3.3-3. Fixed reference channel for maximum input level receiver requirements (SCS 60 kHz, TDD, 64QAM)

Parameter	Unit	Value									
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	100
Subcarrier spacing configuration		2	2	2	2	2	2	2	2	2	2
Allocated resource blocks		11	18	24	31	38	51	65	79	107	135
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		24	24	24	24	24	24	24	24	24	24
MCS Index		24	24	24	24	24	24	24	24	24	24
MCS Table for TBS determination		64QAM									
Modulation		64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM	64 QAM
Target Coding Rate		3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4	3/4
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot											
For Slots 0,1,2,3 and Slot i, if mod(i, 20) = {14,15,16,17,18,19} for i from {0,...,39}	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 20) = {0,..., 13} for i from {4,...,39}	Bits	5376	8712	11784	15112	18432	25104	31752	38936	52224	65576
Transport block CRC	Bits	24	24	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot											
For Slots 0,1,2,3 and Slot i, if mod(i, 20) = {14,15,16,17,18,19} for i from {0,...,39}	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 20) = {0,..., 13} for i from {4,...,39}	CBs	1	2	2	2	3	3	4	5	7	8
Binary Channel Bits per Slot											
For Slots 0,1,2,3 and Slot i, if mod(i, 20) = {14,15,16,17,18,19} for i from {0,...,39}	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 20) = {0,..., 13} for i from {4,...,39}	Bits	7128	11664	15552	20088	24624	33048	42120	51192	69336	87480
Max. Throughput averaged over 1 frame	Mbps	12.902	20.909	28.282	36.269	44.237	60.250	76.205	93.446	125.338	157.382

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.3.1-1.

NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).

NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame

NOTE 4: Slot i is slot index per frame

A.3.3.4 FRC for maximum input level for 256 QAM

Table A.3.3.4-1 Fixed reference channel for maximum input level receiver requirements (SCS 15 kHz, TDD, 256QAM)

[illegible]

Table A.3.3.4-2 Fixed Reference channel for maximum input level receiver requirements (SCS 30 kHz, TDD, 256QAM)

Parameter	Unit	Value									
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	100
Subcarrier spacing configuration		1	1	1	1	1	1	1	1	1	1
Allocated resource blocks		24	38	51	65	78	106	133	162	217	273
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		11	11	11	11	11	11	11	11	11	11
MCS Index		23	23	23	23	23	23	23	23	23	23
MCS Table for TBS determination		256QAM									
Modulation		256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM
Target Coding Rate		4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot											
For Slots 0,1,2 and Slot i, if $\text{mod}(i, 10) = \{7,8,9\}$ for i from $\{0,...,19\}$	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if $\text{mod}(i, 10) = \{0,1,2,3,4,5,6\}$ for i from $\{3,...,19\}$	Bits	16136	25608	33816	44040	52224	71688	90176	108552	147576	184424
Transport block CRC	Bits	24	24	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot											
For Slots 0,1,2 and Slot i, if $\text{mod}(i, 10) = \{7,8,9\}$ for i from $\{0,...,19\}$	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if $\text{mod}(i, 10) = \{0,1,2,3,4,5,6\}$ for i from $\{3,...,19\}$	CBs	1	1	1	1	1	1	2	2	2	3
Binary Channel Bits per Slot											
For Slots 0,1,2 and Slot i, if $\text{mod}(i, 10) = \{7,8,9\}$ for i from $\{0,...,19\}$	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if $\text{mod}(i, 10) = \{0,1,2,3,4,5,6\}$ for i from $\{3,...,19\}$	Bits	20736	32832	44064	56160	67392	91584	114912	139968	187488	235872
Max. Throughput averaged over 1 frame	Mbps	17.750	28.169	37.198	48.444	57.446	78.857	99.194	119.407	162.334	202.866

NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.3.1-1.

NOTE 2: If more than one Code Block is present, an additional CRC sequence of $L = 24$ Bits is attached to each Code Block (otherwise $L = 0$ Bit).

NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame

NOTE 4: Slot i is slot index per frame

Table A.3.3.4-3 Fixed reference channel for maximum input level receiver requirements (SCS 60 kHz, TDD, 256QAM)

Parameter	Unit	Value									
Channel bandwidth	MHz	10	15	20	25	30	40	50	60	80	100
Subcarrier spacing configuration		2	2	2	2	2	2	2	2	2	2
Allocated resource blocks		11	18	24	31	38	51	65	79	107	135
Subcarriers per resource block		12	12	12	12	12	12	12	12	12	12
Allocated slots per Frame		24	24	24	24	24	24	24	24	24	24
MCS Index		23	23	23	23	23	23	23	23	23	23
MCS Table for TBS determination		256QAM									
Modulation		256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM	256 QAM
Target Coding Rate		4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5	4/5
Maximum number of HARQ transmissions		1	1	1	1	1	1	1	1	1	1
Information Bit Payload per Slot											
For Slots 0,1,2,3 and Slot i, if mod(i, 20) = {14,15,16,17,18,19} for i from {0,...,39}	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 20) = {0,..., 13} for i from {4,...,39}	Bits	7424	12040	16136	21000	25608	33816	44040	53288	71688	90176
Transport block CRC	Bits	24	24	24	24	24	24	24	24	24	24
LDPC base graph		1	1	1	1	1	1	1	1	1	1
Number of Code Blocks per Slot											
For Slots 0,1,2,3 and Slot i, if mod(i, 20) = {14,15,16,17,18,19} for i from {0,...,39}	CBs	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 20) = {0,..., 13} for i from {4,...,39}	CBs	1	2	3	3	4	5	6	7	9	12
Binary Channel Bits per Slot											
For Slots 0,1,2,3 and Slot i, if mod(i, 20) = {14,15,16,17,18,19} for i from {0,...,39}	Bits	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
For Slot i, if mod(i, 20) = {0,..., 13} for i from {4,...,39}	Bits	9504	15552	20736	26784	32832	44064	56160	68256	92448	116640

Max. Throughput averaged over 1 frame	Mbps	17.818	28.896	38.726	50.400	61.459	81.158	105.696	127.891	172.051	216.422
NOTE 1: Additional parameters are specified in Table A.3.1-1 and Table A.3.3.1-1.											
NOTE 2: If more than one Code Block is present, an additional CRC sequence of L = 24 Bits is attached to each Code Block (otherwise L = 0 Bit).											
NOTE 3: SS/PBCH block is transmitted in slot #0 of each frame											
NOTE 4: Slot i is slot index per frame											

A.4 CSI reference measurement channels

A.5 OFDMA Channel Noise Generator (OCNG)

A.5.1 OCNG Patterns for FDD

A.5.1.1 OCNG FDD pattern 1: Generic OCNG FDD Pattern for all unused REs

Table A.5.1.1-1: OP.1 FDD: Generic OCNG FDD Pattern for all unused REs

OCNG Parameters	OCNG Appliance	Control Region (Core Set)	Data Region
Resources allocated		All unused REs (Note 1)	All unused REs (Note 2)
Structure		PDCCH	PDSCH
Content		Uncorrelated pseudo random QPSK modulated data	Uncorrelated pseudo random QPSK modulated data
Transmission scheme for multiple antennas ports transmission		Single Tx port transmission	Spatial multiplexing using any precoding matrix with dimensions same as the precoding matrix for PDSCH
Subcarrier Spacing		Same as for RMC PDCCH in the active BWP	Same as for RMC PDSCH in the active BWP
Power Level		Same as for RMC PDCCH	Same as for RMC PDSCH
NOTE 1: All unused REs in the active CORESETS appointed by the search spaces in use. NOTE 2: Unused available REs refer to REs in PRBs not allocated for any physical channels, CORESETS, synchronization signals or reference signals in channel bandwidth.			

A.5.2 OCNG Patterns for TDD

A.5.2.1 OCNG TDD pattern 1: Generic OCNG TDD Pattern for all unused REs

Table A.5.2.1-1: OP.1 TDD: Generic OCNG TDD Pattern for all unused REs

OCNG Parameters	OCNG Appliance	Control Region (Core Set)	Data Region
-----------------	----------------	------------------------------	-------------

Resources allocated	All unused REs (Note 1)	All unused REs (Note 2)
Structure	PDCCH	PDSCH
Content	Uncorrelated pseudo random QPSK modulated data	Uncorrelated pseudo random QPSK modulated data
Transmission scheme for multiple antennas ports transmission	Single Tx port transmission	Spatial multiplexing using any precoding matrix with dimensions same as the precoding matrix for PDSCH
Subcarrier Spacing	Same as for RMC PDCCH in the active BWP	Same as for RMC PDSCH in the active BWP
Power Level	Same as for RMC PDCCH	Same as for RMC PDSCH
NOTE 1: All unused REs in the active CORESETS appointed by the search spaces in use. NOTE 2: Unused available REs refer to REs in PRBs not allocated for any physical channels, CORESETs, synchronization signals or reference signals in channel bandwidth.		

A.6 Void

Annex B (informative): Void

Annex C (informative): Downlink physical channels

C.1 General

The following clauses, describes the downlink Physical Channels that are transmitted during a connection i.e., when measurements are done.

C.2 Setup

Table C.2-1 describes the downlink Physical Channels that are required for connection set up.

**Table C.2-1: Downlink Physical Channels required
for connection set-up**

Physical Channel
PBCH
SSS
PSS
PDCCH
PDSCH
PBCH DMRS
PDCCH DMRS
PDSCH DMRS
CSI-RS

C.3 Connection

C.3.1 Measurement of Receiver Characteristics

Unless otherwise stated, Table C.3.1-1 is applicable for measurements on the Receiver Characteristics (clause 7).

Table C.3.1-1: Downlink Physical Channels transmitted during a connection (FDD and TDD)

Parameter	Unit	Value
SSS transmit power	W	Test specific
EPRE ratio of PSS to SSS	dB	0
EPRE ratio of PBCH to SSS	dB	0
EPRE ratio of PBCH to PBCH DMRS	dB	0
EPRE ratio of PDCCH to SSS	dB	0
EPRE ratio of PDCCH to PDCCH DMRS	dB	0
EPRE ratio of PDSCH to SSS	dB	0
EPRE ratio of PDSCH to PDSCH DMRS (Note 1)	dB	-3
EPRE ratio of CSI-RS to SSS	dB	0
EPRE ratio of PTRS to PDSCH	dB	Test specific
EPRE ratio of OCNG DMRS to SSS	dB	0
EPRE ratio of OCNG to OCNG DMRS (Note 1)	dB	0
NOTE 1: No boosting is applied to any of the channels except PDSCH DMRS. For PDSCH DMRS, 3 dB power boosting is applied assuming DMRS Type 1 configuration when DMRS and PDSCH are TDM'ed and only half of the DMRS REs are occupied.		
NOTE 2: Number of DMRS CDM groups without data for PDSCH DMRS configuration for OCNG is set to 1.		

Annex D (normative): Characteristics of the interfering signal

D.1 General

Some RF performance requirements for the NR UE receiver are defined with interfering signals present in addition to the wanted signal.

For NR bands with $F_{DL_high} < 2700$ MHz and $F_{UL_high} < 2700$ MHz, a modulated 5 MHz full bandwidth NR down link signal, and in some cases an additional CW signal, are used as interfering signal. For intra-band contiguous CA bandwidth class C, a modulated 5 MHz NR downlink signal is used. And for some cases an additional CW signal is used.

For NR bands with $F_{DL_low} \geq 3300$ MHz and $F_{UL_low} \geq 3300$ MHz, a modulated NR downlink signal which equals to channel bandwidth of the wanted signal for single carrier and inter-band CA cases is used as interfering signal. For intra-band contiguous CA bandwidth Class C, a modulated NR downlink signal which equals to the aggregated channel bandwidth of the wanted signal is used. For intra-band contiguous CA bandwidth class D and E cases, a modulated 50 MHz NR downlink signal is used. And for some cases an additional CW signal is used.

D.2 Void

Annex E (normative): Environmental conditions

E.1 General

This normative annex specifies the environmental requirements of the UE. Within these limits the requirements of the present documents shall be fulfilled.

E.2 Environmental

The requirements in this clause apply to all types of UE(s).

E.2.1 Temperature

The UE shall fulfill all the requirements in the full temperature range of:

Table E.2.1-1: Temperature conditions

+15C to +35C	For normal conditions (with relative humidity up to 75 %)
-10C to +55C	For extreme conditions (see IEC publications 6821 and 6822)

Outside this temperature range the UE, if powered on, shall not make ineffective use of the radio frequency spectrum. In no case shall the UE exceed the transmitted levels as defined in clause 6.2 for extreme operation.

E.2.2 Voltage

The UE shall fulfil all the requirements in the full voltage range, i.e. the voltage range between the extreme voltages.

The manufacturer shall declare the lower and higher extreme voltages and the approximate shutdown voltage. For the equipment that can be operated from one or more of the power sources listed below, the lower extreme voltage shall not be higher, and the higher extreme voltage shall not be lower than that specified below.

Table E.2.2-1: Voltage conditions

Power source	Lower extreme voltage	Higher extreme voltage	Normal conditions voltage
AC mains	0,9 * nominal	1,1 * nominal	nominal
Regulated lead acid battery	0,9 * nominal	1,3 * nominal	1,1 * nominal

Non regulated batteries:			
Leclanché	0,85 * nominal	Nominal	Nominal
Lithium	0,95 * nominal	1,1 * Nominal	1,1 * Nominal
Mercury/nickel & cadmium	0,90 * nominal		Nominal

Outside this voltage range the UE if powered on, shall not make ineffective use of the radio frequency spectrum. In no case shall the UE exceed the transmitted levels as defined in clause 6.2 for extreme operation. In particular, the UE shall inhibit all RF transmissions when the power supply voltage is below the manufacturer declared shutdown voltage.

E.2.3 Vibration

The UE shall fulfil all the requirements when vibrated at the following frequency/amplitudes.

Table E.2.3-1: Vibration conditions

Frequency	ASD (Acceleration Spectral Density) random vibration
5 Hz to 20 Hz	0.96 m ² /s ³
20 Hz to 500 Hz	0.96 m ² /s ³ at 20 Hz, thereafter -3 dB/Octave

Outside the specified frequency range the UE, if powered on, shall not make ineffective use of the radio frequency spectrum. In no case shall the UE exceed the transmitted levels as defined in TS 38.101-1 for extreme operation.

Annex F (normative): Transmit modulation

F.0 General

While measuring the transmit modulation quality of carriers, an existence of the carrier leakage needs to be taken into account indicated by the parameter *txDirectCurrentLocation* in *UplinkTxDirectCurrent* IE.

F.1 Measurement Point

Figure F.1-1 shows the measurement point for the unwanted emission falling into non-allocated RB(s) and the EVM for the allocated RB(s).

Figure F.1-1: EVM measurement points

F.2 Basic Error Vector Magnitude measurement

The EVM is the difference between the ideal waveform and the measured waveform for the allocated RB(s)

where

is a set of modulation symbols with the considered modulation scheme being active within the measurement period,
are the samples of the signal evaluated for the EVM,

is the ideal signal reconstructed by the measurement equipment, and

is the average power of the ideal signal. For normalized modulation symbols is equal to 1.

The basic EVM measurement interval is defined over one slot in the time domain for PUCCH and PUSCH and over one preamble sequence for the PRACH.

F.3 Basic in-band emissions measurement

The in-band emissions are a measure of the interference falling into the non-allocated resources blocks. The in-band emission requirement is evaluated for PUCCH and PUSCH transmissions. The in-band emission requirement is not evaluated for PRACH transmissions.

The in-band emissions are measured as follows

,

where

is a set of OFDM symbols with the considered modulation scheme being active within the measurement period,

is the starting frequency offset between the allocated RB and the measured non-allocated RB (e.g. or for the first adjacent RB),

(resp.) is the lower (resp. upper) edge of the UL UE channel bandwidth,

and are the lower and upper edge of the allocated BW, and

is the frequency domain signal evaluated for in-band emissions as defined in the clause (ii)

The relative in-band emissions are, given by

where

is the number of allocated RBs

The basic in-band emissions measurement interval is defined over one slot in the time domain. When the PUSCH or PUCCH transmission slot is shortened due to multiplexing with SRS, the in-band emissions measurement interval is reduced by one OFDM symbol, accordingly.

In the evaluation of in-band emissions, the timing is set according to , where sample time offsets and are defined in clause F.4.

F.4 Modified signal under test

Implicit in the definition of EVM is an assumption that the receiver is able to compensate a number of transmitter impairments. The DFT-s-OFDM modulated signals or PRACH signal under test is modified and, in the case of DFT-s-OFDM modulated signals, decoded according to:

where

is the time domain samples of the signal under test.

The CP-OFDM modulated signals or PUSCH demodulation reference signal or PUCCH data signal under test is equalised and, in the case of CP-OFDM modulated signals decoded according to:

where

is the time domain samples of the signal under test.

To minimize the error, the signal under test should be modified with respect to a set of parameters following the procedure explained below.

Notation:

is the sample timing difference between the FFT processing window in relation to nominal timing of the ideal signal.

is the RF frequency offset.

is the phase response of the TX chain.

is the amplitude response of the TX chain.

In the following represents the middle sample of the EVM window of length (defined in the next clauses) or the last sample of the first window half if is even.

The EVM analyser shall

- detect the start of each slot and estimate and ,
- determine so that the EVM window of length is centred
 - on the time interval determined by the measured cyclic prefix minus 16κ samples of the considered OFDM symbol for symbol l for subcarrier spacing configuration μ in a subframe, with $l = 0$ or $l = 7 \cdot 2^\mu$ for normal CP, i.e. the first 16κ samples of the CP should not be taken into account for this step. In the determination of the number of excluded samples, a sampling rate of $1/T_c$ is assumed. If a different sampling rate is used, the number of excluded samples is scaled linearly.

- on the measured cyclic prefix of the considered OFDM symbol for all other symbols for normal CP and for symbol 0 to 11 for extended CP.
- on the measured preamble cyclic prefix for the PRACH

To determine the other parameters a sample timing offset equal to Δt is corrected from the signal under test. The EVM analyser shall then

- correct the RF frequency offset for each time slot, and
- apply an FFT of appropriate size. The chosen FFT size shall ensure that in the case of an ideal signal under test, there is no measured inter-subcarrier interference.

The carrier leakage shall be removed from the evaluated signal before calculating the EVM and the in-band emissions; however, the removed relative carrier leakage power also has to satisfy the applicable requirement.

At this stage the allocated RBs shall be separated from the non-allocated RBs. In the case of PUCCH and PUSCH EVM, the signal on the non-allocated RB(s), Δf , is used to evaluate the in-band emissions.

Moreover, the following procedure applies only to the signal on the allocated RB(s).

- In the case of PUCCH and PUSCH, the UL EVM analyzer shall estimate the TX chain equalizer coefficients and Δf used by the ZF equalizer for all subcarriers by time averaging at each signal subcarrier of the amplitude and phase of the reference and data symbols. The time-averaging length is 1 slot. This process creates an average amplitude and phase for each signal subcarrier used by the ZF equalizer. The knowledge of data modulation symbols may be required in this step because the determination of symbols by demodulation is not reliable before signal equalization.
- In the case of PRACH, the UL EVM analyzer shall estimate the TX chain coefficients and Δf used for phase and amplitude correction and are selected so as to minimize the resulting EVM. The TX chain coefficients are not dependent on frequency, i.e. Δf and Δf . The TX chain coefficient are chosen independently for each preamble transmission and for each Δf .

At this stage estimates of Δf , Δf , and Δf are available. Δf is one of the extremities of the window Δf , i.e. can be Δf or Δf , where Δf if Δf is odd and Δf if Δf is even. The EVM analyser shall then

- calculate EVM_l with Δf set to Δf ,
- calculate EVM_h with Δf set to Δf .

F.5 Window length

F.5.1 Timing offset

As a result of using a cyclic prefix, there is a range of, which, at least in the case of perfect Tx signal quality, would give close to minimum error vector magnitude. As a first order approximation, that range should be equal to the length of the cyclic prefix. Any time domain windowing or FIR pulse shaping applied by the transmitter reduces the range within which the error vector is close to its minimum.

F.5.2 Window length

The window length W affects the measured EVM and is expressed as a function of the configured cyclic prefix length. In the case where equalization is present, as with frequency domain EVM computation, the effect of FIR is reduced. This is because the equalization can correct most of the linear distortion introduced by the FIR. However, the time domain windowing effect can't be removed.

F.5.3 Window length for normal CP

Table F.5.3-1, F.5.3-2, F.5.3-3 below specify the EVM window length (W) for normal CP.

Table F.5.3-1: EVM window length for normal CP for NR, FR1, 15 kHz SCS

Channel Bandwidth (MHz)	FFT size	Cyclic prefix length for symbols 16 and 8-13 in FFT samples	EVM window length W	Ratio of W to total CP length for symbols 16 and 8-13 ¹ (%)
5	512	36	18	50
10	1024	72	36	50
15	1536	108	54	50
20	2048	144	72	50
25	2048	144	72	50
30	3072	216	108	50
40	4096	288	144	50
50	4096	288	144	50
NOTE 1: These percentages are informative and apply to a slot's symbols 1 to 6 and 8 to 13. Symbols 0 and 7 have a longer CP and therefore a lower percentage.				

Table F.5.3-2: EVM window length for normal CP for NR, FR1, 30 kHz SCS

Channel Bandwidth (MHz)	FFT size	Cyclic prefix length for symbols 113 in FFT samples	EVM window length W	Ratio of W to total CP length for symbols 113 ¹ (%)
5	256	18	9	50
10	512	36	18	50
15	768	54	27	50
20	1024	72	36	50
25	1024	72	36	50
30	1536	108	54	50
40	2048	144	72	50
50	2048	144	72	50
60	3072	216	108	50
70	3072	216	108	50
80	4096	288	144	50
90	4096	288	144	50
100	4096	288	144	50
NOTE 1: These percentages are informative and apply to a slot's symbols 1 through 13. Symbol 0 has a longer CP and therefore a lower percentage.				

Table F.5.3-3: EVM window length for normal CP for NR (60 kHz SCS)

Channel Bandwidth (MHz)	FFT size	Cyclic prefix length for symbols in FFT samples	EVM window length W	Ratio of W to total CP length ¹ (%)
10	256	18	9	50
15	384	27	14	50
20	512	36	18	50
25	512	36	18	50
30	768	54	27	50
40	1024	72	36	50

50	1024	72	36	50
60	1536	108	54	50
70	1536	108	54	50
80	2048	144	72	50
90	2048	144	72	50
100	2048	144	72	50
NOTE 1: These percentages are informative and apply to all OFDM symbols within subframe except for symbol 0 of slot 0 and slot 2. Symbol 0 of slot 0 and slot 2 may have a longer CP and therefore a lower percentage.				

F.5.4 Window length for Extended CP

Table F.5.4-1 below specifies the EVM window length (W) for extended CP. The number of CP samples excluded from the EVM window is the same as for normal CP length.

Table F.5.4-1: EVM window length for extended CP for NR, FR1, 60 kHz SCS

Channel Bandwidth (MHz)	FFT size	Cyclic prefix length in FFT samples	EVM window length W	Ratio of W to total CP length ¹ (%)
10	256	64	54	84.4
15	384	96	80	83.3
20	512	128	106	82.8
25	512	128	110	85.9
30	768	192	164	85.4
40	1024	256	220	85.9
50	1024	256	220	85.9
60	1536	384	330	85.9
70	1536	384	330	85.9
80	2048	512	440	85.9
90	2048	512	440	85.9
100	2048	512	440	85.9
NOTE 1: These percentages are informative.				

F.5.5 Window length for PRACH

The table below specifies the EVM window length for PRACH preamble formats for $L_{RA}=839$ and .

Table F.5.5-1 EVM window length for PRACH formats for $L_{RA}=839$

Preamble format	Cyclic prefix length N_{CP}	Nominal FFT size ¹	EVM window length W in FFT samples	Ratio of W to CP^2
0	3168	24576	2307	72.8%
1	21024	24576	20163	95.9%
2	4688	24576	3827	81.6%
3	3168	6144	2952	93.2%
NOTE 1: The use of other FFT sizes is possible as long as appropriate scaling of the window length is applied				
NOTE 2: These percentages are informative				

The table below specifies the EVM window length for PRACH preamble formats for $L_{RA}=139$ and where.

Table F.5.5-2 EVM window length for PRACH formats for $L_{RA}=139$

Preamble format	Cyclic prefix length N_{CP}	Nominal FFT size ¹	EVM window length W in FFT samples	Ratio of W to CP^2
A1	2882	20482	1442	50.0%
A2	5762	20482	4322	75.0%
A3	8642	20482	7202	83.3%
B1	2162	20482	722	33.3%
B2	3602	20482	2162	60.0%
B3	5042	20482	3602	71.4%
B4	9362	20482	7922	84.6%
C0	12402	20482	10962	88.4%
C2	20482	20482	19042	93.0%
NOTE 1: The use of other FFT sizes is possible as long as appropriate scaling of the window length is applied				
NOTE 2: These percentages are informative				

F.6 Averaged EVM

The general EVM is averaged over basic EVM measurements for n slots in the time domain.

where n is

for PUCCH, PUSCH.

The EVM requirements shall be tested against the maximum of the RMS average at the window W extremities of the EVM measurements:

Thus E_{EVM} is calculated using in the expressions above and is calculated using .

Thus we get:

The calculation of the EVM for the demodulation reference signal, E_{DMRS} , follows the same procedure as calculating the general EVM, with the exception that the modulation symbol set defined in clause F.2 is restricted to symbols containing uplink demodulation reference signals.

The basic measurements are first averaged over n slots in the time domain to obtain an intermediate average .

In the determination of each $E_{\text{DMRS},i}$, the timing is set to T_{DMRS} if $i = 1$, and it is set to $T_{\text{DMRS}} + (i-1)T_{\text{slot}}$ otherwise, where E_{EVM} and E_{DMRS} are the general average EVM values calculated in the same n slots over which the intermediate average is calculated. Note that in some cases, the general average EVM may be calculated only for the purpose of timing selection for the demodulation reference signal EVM. Then the results are further averaged to get the EVM for the demodulation reference signal, E_{DMRS} .

The PRACH EVM, E_{PRACH} , is averaged over 2 preamble sequence measurements for long preamble formats as defined in table 6.3.3.1-1 in [6] and averaged over 10 preamble sequence measurements for short preamble formats as defined in table 6.3.3.1-2 in [6]. The EVM requirements shall be tested against the maximum of the RMS average at the window W extremities of the EVM measurements:

Thus σ^2 is calculated using σ^2_{noise} and $\sigma^2_{\text{channel}}$ is calculated using σ^2_{noise} .

Thus we get:

F.7 Spectrum Flatness

The data shall be taken from FFT coded data symbols and the demodulation reference symbols of the allocated resource block.

F.8

F.9

F.10 EVM for UL MIMO

F10.1 General

EVM for UL MIMO is measured per layer. A zero-forcing (ZF) MIMO receiver architecture is used so that dual layer transmissions by the UE can be demodulated by the test equipment receiver.

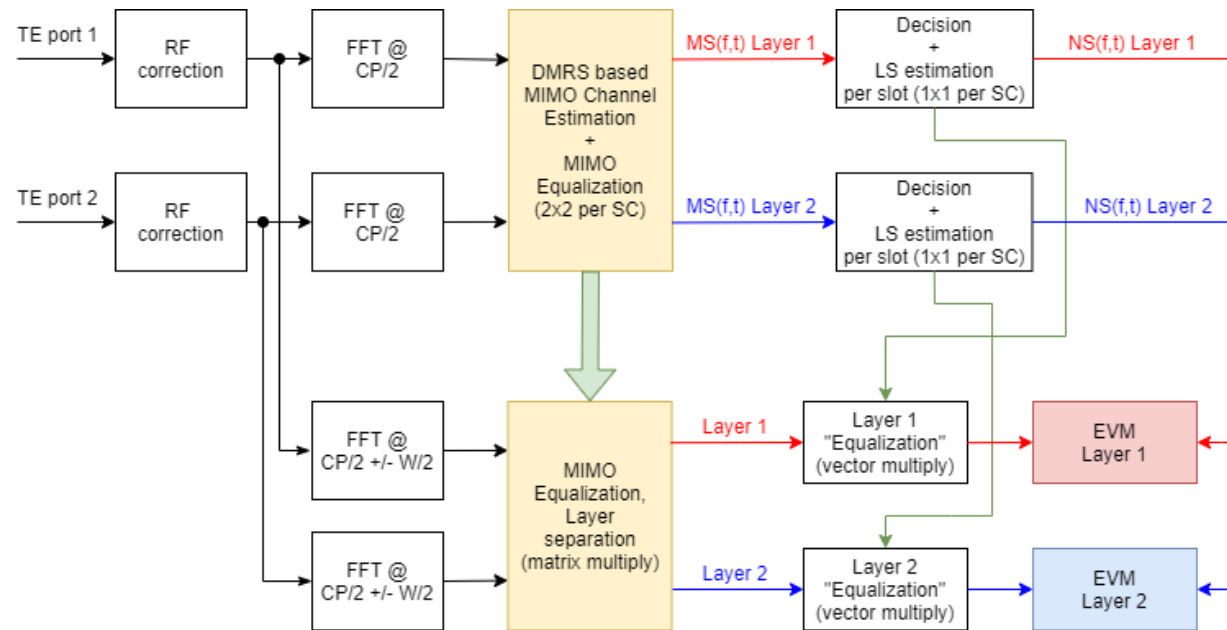


Figure F.10.1-1: EVM calculation block diagram for 2-Layer UL MIMO

The TE receives signals from 2 different ports which are connected to two antenna connectors in the test system. For UL MIMO measurements a MIMO equalization step as described in section F.10.2 is performed to separate the layers. Each layer is then processed as described in section F.10.3 to receive the measurement results for each individual layer.

F10.2 MIMO Equalization

The MIMO equalization is based only on reference signals (DMRS) without using any data symbols. For the equalization process all available DMRS symbols shall be used.

The effective 2x2 channel matrix is estimated using reference signals of different subcarriers, e.g. in case of DMRS antenna ports 0 and 2. In case that same subcarriers are used, e.g. DMRS antenna ports 0 and 1, a channel decomposition is necessary taking advantage of the orthogonal codes w_f and w_t and assuming identical channel coefficients for adjacent subcarriers of same CDM group.

Effective channel including the precoding matrix P is:

with

where y denotes the received symbol on port index n and r the reference signal for layer index v .

Since reference signals of a specific layer are transmitted only on subcarriers of one CDM group channel, interpolation is needed in order to obtain channel coefficients for all subcarriers. Channel interpolation is done using the channel coefficients of active CDM group in all other CDM groups.

The channel coefficients used to calculate the equalizer coefficients are obtained after channel smoothing in frequency domain by computing the moving average of interpolated channel coefficients. The moving average window size is 7. For subcarriers at or near the edge of allocation the window size is reduced accordingly.

The ZF equalizer coefficients are calculated as the inverse of the effective channel matrix, in general:

F10.3 Layer processing

After performing the MIMO equalization as described in section F.10.2 each layer is processed using the existing procedure as defined in Annex E of TS 38.521-1 [4].

Since the channel estimation is calculated only on the DMRS symbols, an averaging including all 14 symbols of one slot, i.e. data and reference signals, is needed in order to minimize EVM. The averaging is achieved by the least square (LS) equalization method described for single layer in Annex E.3. of TS 38.521-1 [4].

$MS(f,t)$ and $NS(f,t)$ are processed with a LS estimator, to derive one equalizer coefficient per time slot and per allocated subcarrier. $EC(f)$ is defined for each layer as:

With $*$ denoting complex conjugation. $EC(f)$ are used to equalize layer data symbols.

EVM equalizer spectral flatness is derived from equalizer coefficients for each layer as follows:

Annex G (normative):

Difference of relative phase and power errors

G.0 General

This annex gives further information needed for understanding and implementing 6.4D.4. The following terms should be understood as follows:

Relative phase error: refers to the phase difference between signals at different antenna connectors, which should be ideally 0. It should be understood as for a slot i.e. (slot) relative phase. It is calculated based on DMRS symbols of that slot or on SRS symbols.

Difference of relative phase error: refers to the difference between the relative phase error determined per slot and the relative phase error determined based on the SRS transmitted.

G.1 Measurement Point

Figure G.1-1 shows the measurement point for the difference of relative phase and power errors.

Figure G.1-1 - Measurement point for difference of relative phase/power error for UL coherent MIMO

G.2 Relative Phase Error Measurement

Here are listed the different aspects that may lead to different interpretations.

G.2.1 Symbols and subcarriers used

Phase error is determined based on DMRS REs (DMRS mapping type A with 3 DMRS symbols per slot, the REs corresponding to the odd subcarriers and DMRS symbols are non-allocated for data or DMRS.) and SRS REs (with 4 SRS symbols in the SRS slot, same SRS resource mapping is used for non-codebook-based and codebook-based precoding).

For the DMRS and SRS to occupy identical SCs and maximize their frequency density, DMRS configuration type 1 and SRS comb2 configuration are used.

UL RMC described in Annex A.2 is used.

G.2.2 CFO (carrier frequency offset) correction

The TE performs a CFO correction on a slot-by-slot basis using a common frequency correction at the two uplink antenna connectors.

G.2.3 Steps of the measurement method

Below are detailed the steps necessary to obtain the maximum difference of relative phase error during the 20ms time window.

- 1 Determination for each subcarrier and at each antenna, the SRS relative phase error based on the last SRS transmitted on Ant1 and Ant2, that relative phase error serves as a reference for the calculation of the difference of relative phase error for each slot inside the 20 ms time window.

The output is the “SRS relative phase error” vector for the last SRS transmitted: .

Calculation for the last SRS transmitted, for each RB of the SRS relative phase errors based on the arithmetic 2mean of the subcarrier SRS relative phase errors determined in previous step.

The output is the “SRS relative phase error” vector for the last SRS transmitted: .

- 3 CFO correction on slot-by-slot basis using a common frequency correction for both antenna outputs.
- 4 Determination for each subcarrier and at each antenna, the phase over the slot being analyzed. The phase is extracted from the channel estimate derived from the 3 DMRS symbols of the slot using the LSE technique.

The output is one vector of dimension for each antenna.

- 5 Calculation for a slot for each subcarrier of the relative phase error (difference between the vectors determined in the previous step).

The output is subcarrier relative phase errors of a slot: .

- 6 Calculation for a slot, for each RB of the relative phase errors based on the arithmetic mean of the subcarrier relative phase errors determined in previous step.

The output is a “slot relative phase error” vector for a slot:.

- 7 Calculation for a slot of the difference of relative phase errors based on the “SRS relative phase error” (reference) determined in step 2 and the “slot relative phase error” determined in previous step.

The output is a “difference of relative phase error” vector for a slot:.

- 8 Calculation for a slot of the arithmetic mean value of the “difference of relative phase error” vector determined in previous step, this value corresponds to an RB.

The output is a “difference of relative phase error” value for a slot:

- 9 Perform for each slot of the 20ms time window, steps 3 to 8.

The output is a “difference of relative phase error” vector: .

- 10 Calculation of the maximum value of the “difference of relative phase error”.

The output is the “difference of relative phase error” that should be verified as complying with the 40° maximum allowable difference of relative phase error requirement: .

Annex H (normative): ModifiedMPR-Behavior

H.1 Indication of modified MPR behavior

This annex contains the definitions of the bits in the field *modifiedMPR-Behavior* indicated per supported NR band in the IE *RF-Parameters* [7] by a UE supporting an MPR or A-MPR modified in a given version of this specification. A modified MPR or A-MPR behaviour can apply to a supported NR band in stand-alone operation (including CA and NN-DC operation) or in non-standalone operation with the said NR band as part of an EN-DC or NE-DC band combination.

NOTE 1: In the present release, the *modifiedMPR-Behavior* is indicated [7] by an 8-bit bitmap per supported NR band.

Table H.1-1: Definitions of the bits in the field *modifiedMPR-Behavior*

NR Band	Index of field (bit number)	Definition (description of the supported functionality if indicator set to one)	Notes
n41	0 (leftmost bit)	- EN-DC contiguous intraband MPR as defined in clause 6.2B.2.1 of 38.101-3 v15.5.0	- This bit may be set to 1 by a UE supporting DC_(n)41AA UE EN-DC
	1	- EN-DC non-contiguous intraband MPR as defined in clause 6.2B.2.2 of 38.101-3 v15.5.0	- This bit may be set to 1 by a UE supporting DC_41A_n41A EN-DC
n71	0 (leftmost bit)	- EN-DC contiguous intraband MPR as defined in clause 6.2B.2.1 of 38.101-3 v15.5.0	- This bit may be set to 1 by a UE supporting DC_(n)71AA UE EN-DC

Annex I (informative): Void

Annex J (informative): Void

Annex K (informative): Void

Annex L (informative): Change history

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2017-08	RAN4#84	R4-170890 9				Initial Skeleton	0.0.1
2017-10	RAN4#84 Bis	R4-170995 8				Added approved TPs in RAN4-NR-AH#3 R4-1709948, TP for TS 38.101-1: minimum output power, Huawei R4-1709454, TP for TS 38.101-1:UE Tx spurious emission for range 1, ZTE Corporation	0.1.0
2017-10	RAN4#84 Bis	R4-1711978				Embedded approved TPs in RAN4#84Bis R4-1711556, "TP to TS 38.101: Draft CR to Transmitter power clause", Nokia R4-1710962, "TP to TS 38.101-1: Draft CR to Output RF spectrum emissions" Nokia R4-1711608, "TP for TS38.101-1 on conducted UE transmitter intermodulation for FR1(section 6.5)" ZTE Corporation Number of TPs by editors	0.2.0
2017-12	RAN4#85	R4-171380 5				Approved TPs in RAN4#85 R4-1713204, TP on general parts for 38.101-1 NR FR1, Ericsson R4-1714047, WF on MPR for sub6GHz, NTT DOCOMO, INC. R4-1714052, TP for TS 38.101-1 introduction of band n71 for transmitter characteristics, T-Mobile USA Inc. R4-1714162, TP to 38.101-1: ACS, Ericsson R4-1714163, TP to 36.101-1: In-band blocking, Ericsson R4-1714446, TP to 36.101-1: Out-of-band blocking and exceptions for spurious response, Ericsson R4-1714369, TP for NBB requirement for FR1, Intel Corporation R4-1714529, TP on introducing operating bands for NR-LTE DC including SUL band combinations in 38.101-1, Huawei R4-1714097, TP for TS 38.101-1: UE RF requirements for standalone SUL, Huawei R4-1714536, TP for TS 38.101-1: Channel Bandwidth Definition, Qualcomm Incorporated (Note, this TP was further discussed and edited in the reflector) R4-1714114, TP for TS 38.101-1: Channel Arrangement, Qualcomm Incorporated (Note, this TP was further discussed and edited in the reflector) R4-1714029, Sub6 Reference Sensitivity, Qualcomm Incorporated R4-1714329, TP to TR 38.101-01 v0.2.0: ON/OFF mask design for NR UE transmissions for FR1, Ericsson Band list according to R4-1714542, List of bands and band	0.3.0

					combinations to be introduced into RAN4 NR core requirements by December 2017, RAN4 Chairmen Input from: R4-1714479, TP for TR 38.817-01 NR channel bandwidth, Huawei, HiSilicon	
2017-12	RAN4#85	R4-1714569			Further corrections and alignments with 38.104 after email review	0.4.0
2017-12	RAN#78	RP-172475			v1.0.0 submitted for plenary approval. Contents same as 0.4.0	1.0.0
2017-12	RAN#78				Approved by plenary – Rel-15 spec under change control	15.0.0
2018-03	RAN#79	RP-180264	0003		F Implementation of endorsed CRs to 38.101-1 Endorsed draft CRs F: R4-1800400, Editorial corrections for 38.101-1, Qualcomm B: R4-1801102, Draft CR for 30 MHz CBW support, Huawei F: R4-1800032, 38.101-1 n71 draft CR for section 6.2.3 - UE A-MPR - NS values, T-Mobile USA Inc. B: R4-1801121, Draft pCR for TS 38.101-1 version 15.0.0: Remaining ON/OFF masks for FR1 NR UE transmissions, Ericsson F: R4-1800417, Correction of NR SEM table and additional requirements table, vivo F: R4-1800033, 38.101-1 n71 draft CR for section 6.5.3.2 Spurious emissions for UE co-existence, T-Mobile USA Inc. F: R4-1801114, Proposal on protected band numbering in UE specs, Sprint Corporation F: R4-1800407, Draft CR for TS 38.101-1: Mandatory 4Rx antenna performance for NR UE, Vodafone Group Plc F: R4-1800451 Draft CR for TS 38.101-1: Clarification of 4Rx NR bands, Huawei, HiSilicon F: R4-1801136, Draft CR for TS 38.101-1: REFSENS for NR bands, Huawei, HiSilicon F: R4-1801137, Draft CR: n71 REFSENS, Dish Network F: R4-1800395, Draft CR to 38.101-1: corrections to ACS and in-band blocking, Ericsson F: R4-1800396, Draft CR to 38.101-1: corrections to out-of-band blocking, Ericsson F: R4-1800397, Draft CR to 38.101-1: corrections to spurious response, Ericsson F: R4-1800305, Draft CR for NR FR1 wide band intermodulation requirements, MediaTek Inc. F: R4-1800320, Draft CR to 38.101-1: Rx Spurious emission for NR FR1 (section 7.9), ZTE Corporation F: R4-1800473, Draft CR on UE RF requirements for SUL in TS 38.101-1, Huawei F: R4-1800965, Draft CR to TS 38.101-1: Asymmetric CH BW operation, Dish Network F: R4-1800882, Draft CR for correction of UE channel bandwidth for Bands n77 and n78 for TS 38.101-1, Orange UK F: R4-1801012, Draft CR to 38.101-1: Clarifications to UE spectrum utilization section 5.3, Ericsson F: R4-1800030, 38.101-1 n71 draft CR for section 5.4.4 - TX–RX frequency separation, T-Mobile USA Inc F: R4-1801228, Draft CR to 38.101-1: Channel spacing for CA for	15.1.0

					NR FR1(section 5.4.1.2), ZTE Corporation F: R4-1801231, Correction CR for channel spacing:38.101-1, Samsung F: R4-1801235, Draft CR to TS 38.101-1: Corrections on channel raster calculation in section 5.4.2, ZTE Corporation F: R4-1801318, Draft CR on synchronization raster, Huawei RAN4#86: R4-1803053, Draft CR for new spec structure of 38.101-1, Ericsson R4-1801479, Draft CR to 38.101-1: Default Tx-RX frequency separation for NR FR1(section 5.4.4), ZTE R4-1801581, Draft CR for TS 38.101-1 update of 4Rx bands, Huawei Technologies France R4-1802211, draft CR TS 38.101-1 Uplink configuration for FR1 NR REFSENS, Skyworks Solutions Inc. R4-1802342, Draft CR for NR FR1 ACS case 2 transmitter power setting correction (Note 1), MediaTek Inc. R4-1802509, Draft CR on 38.101-1 v15.0.0: Remaining ON/OFF masks for FR1 NR UE transmissions, Ericsson R4-1802566, Draft CR to TS 38.101-1: Clarification of mixed numerology guardband size, Ericsson R4-1802978, Draft CR to TS 38.101-1: Corrections on channel raster in Section 5.4.2.3, Intel Corporation R4-1803064, Draft CR for 38.101-1: Correction of errors, Sprint Corporation R4-1803065, Draft CR for 38.101-1 Introduction of n41requirements, Sprint Corporation R4-1803242, Draft CR to 38.101-1: Corrections to n66, Dish Network R4-1803285, Draft CR to 38.101-1: Correction to CH BWs without symmetric uplink Dish Network, Skyworks Solutions Inc. R4-1803436, Introduction of UL subcarrier alignment for additional bands, AT&T R4-1803456, Draft CR for 38.101-1: Spurious Emissions for UE Coexistence, Sprint Corporation R4-1803461, CR on configured transmitted power for TS 38.101-1, Huawei R4-1803452, draft CR for introduction of completed band combinations from 37.865-01-01 into 38.101-1, Ericsson R4-1803567, Draft CR for TS 38.101-1: Sync raster offset in re-farming bands (5.4.3), Ericsson R4-1803365, CR to introduce MPR for PC2 and PC3 and A-MPR for UTRA protection, Nokia	
2018-06	RAN#80	RP-181262	0011		F CR to TS 38.101-1: Implementation of endorsed draft CRs from RAN4 #86bis and RAN4 #87 R4-1803900, Draft CR into TS 38.101-1 Introduction of band combinations for SUL, Huawei R4-1804021 CR for clarifications for NR FR1 CA BW Classes Nokia, Nokia Shanghai Bell R4-1804140 CR for Narrow Band Blocking requirement for FR1 Intel Corporation	15.2.0

					R4-1804219 Draft CR for 38.101-1: n41 SEM and additional spurious emissions SPRINT Corporation R4-1804266 Draft CR to 38.101-1 MPR channel bandwidth criteria Skyworks Solutions Inc. R4-1804267 Draft CR to 38.101-1 n3,n5,n8 REFSSENS levels Skyworks Solutions Inc. R4-1804268 Draft CR to 38.101-1: Correction to n41 uplink configuration for reference sensitivity Skyworks Solutions Inc. R4-1804370 Draft CR to add missing NR inter-band DL CA in FR1 for TS 38.101-1 NTT DOCOMO, INC. R4-1804581 Draft CR to 38.101-1: On EVM Wording Qualcomm, Inc. R4-1804948 Corrections to 5.3.3 in TS 38.101-1 Nokia, Nokia Shanghai Bell R4-1804877 draft CR introduction completed band combinations 37.865-01-01 -> 38.101-1 Ericsson R4-1805444 Draft CR to TS 38.101-1: Asymmetric CH BW operation Dish Network R4-1805447 drfat CR for including SRS antenna switching in configured output power Qualcomm Incorporated R4-1805462 Editorial corrections to UE RF requirements in 38.101-1 Qualcomm Incorporated R4-1805659 Draft CR for CBW for n50 for 38.101-1 Huawei R4-1805664 Draft CR to 38.101-1: Addition of Annex F Rohde & Schwarz R4-1805665 Correction to inner and outer definitions for MPR Qualcomm Incorporated R4-1805684 Draft CR to TS38.101-1: Channel Raster to Resource Element Mapping (Section 5.4.2.2) and RB alignment with different numerologies (Section 5.3.4) ZTE Corporation R4-1805698 Draft CR for 38.101-1 for Rx(Ch7) of Band n77, n78 and n79 RF requirements CMCC R4-1805699 Draft CR to 38.101-1:introduction of Tx/Rx requirements for inter-band CA ZTE Corporation R4-1805751 Draft CR on UE-to-UE coexistence requirements to protect band 29 from NR band 71 LG Electronics France R4-1805783 Draft CR for 38.101-1 for Tx(Ch6) of Band n77, n78 and n79 RF requirements CMCC R4-1805902 Draft CR into TS 38.101-1 Correction on SUL_n78-n80 Huawei, HiSilicon R4-1805904 Draft CR into TS 38.101-1 Introduction of new band combinations for SUL Huawei, HiSilicon R4-1805921 Draft CR on NR UE REFSSENS SNR FRC for FR1 Intel Corporation R4-1805981 Draft CR for TS38.101-1:Sync raster Samsung R4-1804548 Draft CR for CA BW class for FR1 NTT DOCOMO, INC. R4-1806170 Draft CR on frequency error for TS 38.101-1 ZTE Corporation R4-1806481 Draft CR for Environmental conditions in TS 38.101-1 Annex NTT DOCOMO, INC.	
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					R4-1806657 Draft CR to 38.101-1: Measurement BW for min and off power Skyworks Solutions Inc. R4-1806669 Draft CR to TS38.101-1_introduction of completed band combinations for inter-band 2UL CA ZTE Corporation R4-1806673 Draft CR to TS38.101-1_Remove brackets from Tx and Rx spurious emission table ZTE Corporation R4-1806677 Draft CR on including CA bandwidth class and band combinations for intra-band CA LG Electronics France R4-1806719 Introduction of 7.5 kHz frequency shift for Band n71 Ericsson, T-Mobile R4-1806844 Draft CR for 38.101-1 for Tx(Ch6): missing maximum power requirements for n1 and n8 SoftBank Corp. R4-1806945 Draft CR for TS 38.101-1: Channel raster and NR-ARFCN clarification (5.4.2) Ericsson R4-1807039 Intra-band CA terminology for UE ZTE Corporation R4-1807178 Corrections to n70 TX/RX frequency separation Dish Network R4-1807181 Corrections to spurious emissions UE co-existence table Dish Network R4-1807234 Draft CR into TS 38.101-1 Some Corrections for SUL Huawei, HiSilicon R4-1807269 Corrections to Wide band intermodulation table <2700MHz Dish Network R4-1807392 to remove the brackets for SU in 38.101-1 Huawei, HiSilicon R4-1807647 Draft CR to TS 38.101-1: Correction to Asymmetric CH BW operation Dish Network R4-1807680 Draft CR on 38.101-1 on channel raster to achieve alignment of data and SSB subcarrier grids Nokia, Nokia Shanghai Bell, Ericsson R4-1807705 CR for TS 38.101-1 A-MPR for n51 Huawei, Hisilicon R4-1807814 Draft CR for 38.101-1: SEM correction for n41 Sprint Corporation R4-1807851 Draft CR for 38.101-1: UE spurious emission protection requirements for n5 Sprint Corporation R4-1807920 General requirements for downlink inter-band CA Qualcomm Incorporated R4-1807923 Resolution bandwidth for ACLR Qualcomm Incorporated R4-1808084 Introduction of n12 into TS 38.101-1 Nokia R4-1808087 Draft CR 38.101-1: Introduction of n2, n25, n66 and n70 Sprint Corporation, Dishnetwork R4-1808090 Draft CR to TS 38.101-1: Inclusion of Simultaneous RxTx UE capability for some band combinations Ericsson, Vodafone, Orange R4-1808107 Draft CR to TS38.101-1_corrections on UE coexistence ZTE Corporation R4-1808111 TP to TS38.101-1 - UE ON/OFF masks Ericsson R4-1808116 Draft CR to 38.101-1: introduction of Band n34,n39 and n40 RF requirements ZTE Corporation,CMCC R4-1808136 Draft CR to 38.101-1: FR1 UE Power Control	
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					Qualcomm Incorporated R4-1808141 Correction to MPR for PC2 and spectrum emission mask measurement bandwidth Qualcomm Incorporated R4-1808142 Draft CR for 38.101-1 n41 A-MPR Sprint Corporation, Nokia, Nokia Shanghai Bell, Ericsson R4-1808143 Draft CR for TS 38.101-1 A-MPR for n20 Huawei, HiSilicon R4-1808155 Draft CR for TS 38.101-1: to correct requirements for n71 Samsung R4-1808178 Addition parameters about n50 & n51 in TS 38.101-1 Huawei, Hisilicon, Etisalat (editors note: n50 not implemented per chairmans agreement) R4-1808182 Draft CR for TS 38.101-1 A-MPR for n28 Huawei, HiSilicon R4-1808187 CR for RF requirements for Coherent UL MIMO for FR1 Qualcomm Austria RFFE GmbH R4-1808207 Draft CR to 38.101-1: On EVM Averaging Length, Wording , Qualcomm R4-1808209 Draft CR for 38.101-1 for Tx (Ch6) of HPUE Qualcomm R4-1808466 Draft CR on UL RMC and OCNG pattern for FDD REFSSENS tests RD session R4-1808493 Draft CR for TS 38.101-1: Channel and sync raster corrections (5.4) Ericsson R4-1808507 Draft CR for TS38.101-1 on addition of new 90MHz UE CBW for n41/n78 LG Electronics Inc., LG Uplus, Samsung R4-1808176, Draft CR for 38.101-1 : Introduction of A-MPR for n8, SoftBank R4-1808201, Draft CR for 38.101-1 : Introduction of A-MPR for n1, SoftBank R4-1807101, draft CR introduction completed band combinations 37.865-01-01 -> 38.101-1, Ericsson	
2018-09	RAN#81	RP-181896	0025	F	Big CR for 38.101-1 Endorced draft CRs from RAN4#NR-AH-1807 R4-1809335, Draft CR on UL RMC for FR1 RF tests , Qualcomm Incorporated R4-1809337, Draft CR on NR UE REFSSENS SNR FRC for FR1, Intel Corporation R4-1809339, Draft CR on measurement of receiver characteristics for FR1 RF Tests, Qualcomm Incorporated R4-1809396, Draft CR on NR UE maximum input level FRC for FR1, Intel R4-1809567, Draft CR on OCNG pattern for FR1 REFSSENS tests, Qualcomm Incorporated, Rohde & Schwarz Endorced draft CRs from RAN4#88 R4-1809714, Draft CR to correct in-band blocking parameters for FR1, Anritsu Corporation	15.3.0

					R4-1809784, Draft CR to 38.101-1: Corrections on CA bandwidth classes for FR1, ZTE Corporation R4-1809785, Draft CR to TS 38.101-1 for Corrections on UE transmitter power, ZTE Corporation R4-1809793, Draft CR to 38.101-1: Corrections on additional spectrum emission mask, ZTE Corporation R4-1809919, Correction on UE receiver requirement for FR1, CATT R4-1810091, Draft CR TS 38.101-1 - UE ON-OFF mask clean up, Ericsson R4-1810210, Draft CR for TS 38.101-1: MPR inner and outer RB allocations formula correction, MediaTek, Inc. R4-1810229, Draft CR for TS 38.101-1: Spurious emission for UE coexistence table corrections, MediaTek, Inc. R4-1810230, Draft CR for TS38.101-1 to correct 90MHz UE CBW, LG Electronics, Inc. R4-1810232, Draft CR for TS 38.101-1: Table 7.3.2-1 n77 reference sensitivity corrections, MediaTek, Inc. R4-1810369, Draft CR to 38.101-1: Corrections on symbols and abbreviations in section 3, ZTE Corporation R4-1810376, Draft CR: General corrections to n71 requirements, Dish Network R4-1810428, Draft CR on TS38.101-1 for UE maximum output power for UL MIMO, OPPO R4-1810552, Correction of reference tables, OPPO R4-1810729, Draft CR for introduction of Band n74 for TS 38.101-1, NTT DOCOMO, Inc. R4-1810862, Draft CR to 38.101-1: Updates to Transmit Modulation Annex, Rohde & Schwarz R4-1810892, CR to update Table 6.2D.1-2 for FR1, Qualcomm Incorporated R4-1810961, CR on ACS minimum requirement, Intel Corporation R4-1810965, CR on Out-of-Band Blocking minimum requirement, Intel Corporation R4-1810967, CR on Rx Intermodulation characteristics for CA, Intel Corporation R4-1810974, Annex lettering change for 38.101-1, Qualcomm Incorporated R4-1811189, CR to add more details to Coherent UL MIMO spec for FR1, Qualcomm Incorporated R4-1811280, Corrections of NR receiver characteristics titles, Vivo R4-1811455, Draft CR on DL Physical Channel for FR1 RF tests, Qualcomm Europe Inc. (Spain) R4-1811457, NS numbering, Qualcomm Incorporated R4-1811459, Correction on UE transmitter requirement for FR1, CATT R4-1811463, Draft CR for 38.101-1: Addition of missing NR CA configurations n8-n75 and n28-n75, VodafoneItalia SpA R4-1811472, Addition parameters about n51 in TS 38.101-1, Huawei, Hisilicon, Etisalat R4-1811474, CR CP- OFDM almost contiguous allocation, Nokia, Nokia Shanghai Bell R4-1811477, Draft CR to 38.101-1: FR1 Power Control, Qualcomm	
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						Incorporated R4-1811478, A-MPR correction for n20 and n28, Huawei, HiSilicon R4-1811490, Draft CR to 38.101-1: Addition of Carrier Leakage table, Rohde & Schwarz R4-1811491, Draft CR for TS38.101-1 on transmit signal quality, OPPO R4-1811493, CR to TS 38.101-1: pi/2 BPSK with Spectrum Shaping, Indian Institute of Tech (M), Indian Institute of Tech (H), CEWiT, Nokia R4-1811513, A proposal on 2UL co-ex table modification, SoftBank Corp. R4-1811514, Draft CR to TS 38.101-1: Clarification on OCNG, Keysight Technologies UK Ltd R4-1811516, Draft CR on NR DL FRCs for FR1 UE RF requirements, Intel Corporation R4-1811550, Draft CR to TS 38.101-1 on channel bandwidth and spacing descriptions, Ericsson R4-1811553, Draft CR to 38.101-1: Corrections on description of channel raster entries, ZTE Corporation R4-1811783, Measurement period of PRACH time mask, CATT R4-1811792, Draft CR for A-MPR revision for n1, NTT DOCOMO, INC. R4-1811798, Draft CR for P_{cm}ax for FR1, Qualcomm Incorporated R4-1811799, P_{cm}ax for inter-band NR CA FR1 draft CR, InterDigital, Inc. R4-1811812, Draft CR to 38.101-1: On FR1 AMPR Band n41 NS_04, Qualcomm Incorporated R4-1811816, CR to update the definition of Long and Short subslot for FR1, Qualcomm R4-1811894, Addition parameters about n50 in TS 38.101-1, Huawei R4-1811896, Draft CR for TS 38.101-1: n41 GSCN range modification, MediaTek Inc. R4-1811285, Draft CR TS 38.101-1: NS_04 A-MPR' and spurious emission corrections, Sprint	
2018-12	RAN#82	RP-182836	0029	1	F	Endorced draft CRs from RAN4#88Bis: R4-1812050, CR Simplification of NR NS_08, Nokia R4-1812054, Correction for Inter-band CA operating bands table in TS 38.101-1, Nokia. R4-1812079, draft CR to introduce asymmetric UL DL channel BW combinations for n71, T-Mobile USA Inc. R4-1812121, Draft CR on Note1 Corrections in 38.101 RX tests, Qualcomm R4-1812128, draftCR on 256QAM UL power requirement, Intel Corporation R4-1812200, Draft CR to TS 38.101-1 Add clarification note to PC3 MPR table, Intel Corporation R4-1812217, Draft CR to 38.101-1: Corrections on the descriptions of UE channel bandwidth for CA, ZTE Corporation R4-1812319, Draft CR for TS 38.101-1: REFSENS UL configuration	15.4.0

					corrections, MediaTek Inc. R4-1812320, Draft CR for TS 38.101-1: Out-of-band blocking exceptions for CA, MediaTek Inc. R4-1812322, Draft CR for TS 38.101-1: Blocking characteristics for SUL, MediaTek Inc. R4-1812397, Clarification for almost contiguous CP-OFDM, Qualcomm Incorporated R4-1812508, Draft CR to 38.101-1: Corrections on channel raster & SS raster for operating bands, ZTE Corporation R4-1812611, Draft CR to 38.101-1: Some corrections for inter-band CA combinations, ZTE Corporation R4-1813459, Draft CR for TS 38.101-1: Support 4Rx for n38, Huawei R4-1813469, draftCR on applicability of TDD configuratiin for CA in TS 38.101-1, Huawei R4-1813521, Addition of ?TC,c for single carrier P_{max} for FR1, vivo R4-1813798, Draft CR to 38.101-1: Corrections on UE additional maximum output power reduction, ZTE Corporation R4-1813811, Draft CR to 38.101-1: Correction to n12 reference sensitivity power levels, Skyworks Solutions Inc. R4-1813812, Band n41 spurious emission limits , Qualcomm Incorporated R4-1813813, Draft CR for TS 38.101-1: P-Max for 5G NR HPUE, CMCC R4-1814158, CR on Spurious emissions for UE co-existence, Intel Corporation R4-1814159, Draft CR for CA ACS/IBB for Bandwidth class C, Qualcomm R4-1813843, Draft CR to 38.101-1: Update of Annex F, Rohde & Schwarz R4-1813845, Correction for PI/2 PSK requiements, Nokia Endorsed draft CR's from RAN4#89 R4-1815950, dCR on TS38.101-1 merging draft CRs from RAN4#88Bis, Qualcomm Incorporated R4-1814752, DraftCR to TS 38.101-1 pi/2 BPSK in n41, CMCC R4-1814824, n50 A-MPR, Qualcomm Incorporated R4-1814959, Changes to Max input power UL and DL configuratgions in FR1, OPPO R4-1814970, NR FR1 relative power tolerance CR, Nokia R4-1814972, A-MPR for NS_03 and NS_03U and re-formulation of NS_100, Nokia R4-1815060, draft CR for adding note about the fallback of NR CA in FR1 for TS 38.101-1, NTT DOCOMO, INC. R4-1815392, Draft CR to 38.101-1: Update to NS_04 requirements, Rohde & Schwarz R4-1815563, Draft CR to 38.101-1 on Clarification on 7.5 KHz raster shift in NR re-farmed bands, Ericsson R4-1815863, Draft CR for 38.101-1: Nominal carrier spacing for 30 kHz raster, SPRINT Corporation R4-1815898, draft CR on CA configuration on bandwidth class F, Huawei R4-1815917, draftCR on DL RMC for TS 38.101-1, Huawei	
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						R4-1816162, Draft CR on introduction of SRS switch IL in FR1, OPPO R4-1816199, Draft CR on FR1-FR2 UE-to-UE coexistence for TS38.101-1, LG Electronics France R4-1816200, Draft CR to 38.101-1 on intra-band contiguous CA configurations for FR1, ZTE Corporation R4-1816240, Transient period for SRS Antenna Switching for FR1, Qualcomm R4-1816243, Draft CR to TS38.101-1 Clarifications on MSD and UL configuration tables for inter-band CA, ZTE Corporation R4-1816466, Draft CR on some changes for SUL band combinations to TS 38.101-1, Huawei R4-1816468, Support of 7.5 kHz carrier shift for additional operating bands, Ericsson R4-1816604, TDD configuration for UE Tx test in FR1, Ericsson R4-1816663, Draft CR to 38.101-1 (5.3.4) RB alignment, Huawei R4-1816755, CR to 38.101-1: ACS and IBB intra-band contiguous CA, Intel Corporation Further changes in RAN#82 - 7.5 kHz frequency shift is specified for all FDD bands in clause 5.4.2.1	
2018-12	RAN#82	RP-182814	0030	2	F	Company CR on 2Rx exception for NR vehicular UE at FR1	15.4.0
2019-03	RAN#83	RP-190403	0034		F	CR to TS 38.101-1: Implementation of endorsed draft CRs from RAN4#90 Endorsed draft CR from Ran4#90 R4-1900032, Editorial corrections for 38.101-1, Qualcomm Incorporated R4-1900031, draftCR on SRS IL for CA, Qualcomm Incorporated R4-1900161, CR on Relative power tolerance, Intel Corporation R4-1900162, CR on Minimum output power, Intel Corporation R4-1900274, Draft CR to TS 38.101-1 on NR general spectrum emission mask, ZTE Corporation R4-1900275, Draft CR to TS 38.101-1 on spurious emission for network signalled value NS_40, NS_41 and NS_42, ZTE Corporation R4-1900424, Correction of table references and other typos, Ericsson R4-1900508, Draft CR to TS 38.101-1 on UE transmitter power and some other editorial corrections, ZTE Corporation R4-1900723, Draft CR on editorial error of TS38.101-1, LG Electronics Inc. R4-1900727, Update to PRACH EVM window length for FR1, Rohde & Schwarz R4-1900840, Draft CR for 38.101-1 modification of Transmit intermodulation requirement, Huawei R4-1900848, [RAN5 LS]Draft CR for 38.101-1: adding note for inter-band CA spurious emissions, Huawei R4-1901033, Alignment of Foob related description for 38.101-1, vivo	15.5.0

						R4-1901273, Correction of HARQ-ACK transmission timing for DL RMC for FR1 TDD SCS=60kHz, Ericsson R4-1901766, draft_CR TS 38.101-1 Correction to UL configuration for reference sensitivity, Skyworks Solutions Inc. R4-1901823, draft CR on spurious requirment for TS 38.101-1, Huawei, HiSilicon R4-1901835, draftCR on MSD for CA_n41-n78 for TS 38.101-1, Huawei R4-1901847, Draft CR for 38.101-1: Addition of default power class, Sprint Corporation R4-1901873, Receiver requirement RMC references, Qualcomm Incorporated R4-1901925, Draft CR to 38.101-1 to update and clarify Rx wide band intermod and spurious requirments for BW class C, D, E, Qualcomm Incorporated R4-1901992, Draft CR to 38.101-1. Correct FR1 NS_41 AMPR for n50, Huawei R4-1902001, Draft CR to 38.101-1 on n41 – B40 coexistence, Qualcomm Incorporated R4-1902150, Draft CR to TS38.101-1_Clarifications on MSD and UL configuration tables for inter-band CA, ZTE Corporation R4-1902166, Tx ON/OFF time mask for FR1, Qualcomm Inc R4-1902174, Draft CR to 38.101-1: On FR1 A-MPR NS_08 for n8, Qualcomm Incorporated R4-1902175, Draft CR on AMPR requirements for NS_05U and NS_08U to TS 38.101-1, Huawei R4-1902194, [41 DL]Draft CR for 38.101-1 adding DL intra-band CA requirements for frequency less than 2700MHz, Huawei R4-1902196, Draft CR for 7.9A Spurious emissions for CA, CMCC R4-1902223, UE optional bandwidth for FR1, Nokia R4-1902225, CR to 38.101-1 on CA BW Classes fallback groups, Intel Corporation R4-1902233, Draft CR to 38.101-1: SUL clarifications, Nokia R4-1902339, Draft CR to TS 38.101-1 on FR1 extension, Ericsson R4-1902455, Completion of the P_{cm}ax specification: additional P_{max} and P_{NR}, Ericsson R4-1902468, Draft CR: Introduction of Annex on Characteristics of the Interfering Signal, Samsung R4-1902479, Draft CR on some errors to TS 38.101-1, Huawei R4-1902480, Draft CR for 38.101-1 modification of requirements for network signalled value NS_04, Huawei R4-1902655, CR to 38.101-1 on NR Uplink RBs location, Intel Corporation R4-1901610, Draft CR for 38.101-1 REFSSENS for UL MIMO, Huawei Editorial changes after RAN#83 To align the annex numbering with other specifications (TS 38.101-x series), annexes J and K were added and Change history was numbered as annex L.	
2019-06	RAN#84	RP-191240	0047		F	CR to TS 38.101-1: Implementation of endorsed draft CRs from RAN4#90bis and RAN4#91	15.6.0

					Endorced draft CRs from RAN4#90Bis R4-1902826, Draft CR for 38.101-1 modification of ACS test parameters case 2 for intra-band contiguous CA , Huawei R4-1902926, Draft CR to TS 38.101-1 Correction to Pcmx, Intel Corporation R4-1902975, Draft CR on PRACH and PUCCH format description for EVM in FR1, Anritsu corporation R4-1903032, Draft CR on editorial error of TS38.101-1, LG Electronics France R4-1903120, Draft CR on DL power allocation for TS 38.101-1, Intel Corporation R4-1903124, Draft CR on b41-n40 coexistence, Intel Corporation R4-1903151, Draft CR to TS38.101-1_removing DC sections, ZTE Corporation R4-1903195, Draft CR for 38.101-1: remove the bracket of UE capability "powerBoosting-pi2BPSK", Huawei R4-1903392, Draft CR for TS 38.101-1: Corrections to EVM equalizer spectrum flatness requirements, MediaTek Inc. R4-1903473, Draft CR on FREF,Shift, CMCC R4-1903508, Draft CR to TS 38.101-1 on spurious emissions for UE co-existence, ZTE Corporation R4-1904335, DraftCR TS 38.101 Corrections to NS_100 UTRA ACLR frequency band list, Skyworks Solutions Inc. R4-1904460, Draft CR for 38.101-1 CA Pcmx, Huawei R4-1904537, Draft CR for TR 38.101-1 correction of A-MPR for NS_04, Huawei R4-1904554, Draft CR to 38.101-1: FR1 power dynamics DTX removal, Qualcomm Incorporated R4-1904927, Draft CR to clarify frequency of carrier leakage in RBs for FR1, Anritsu corporation R4-1904928, Draft CR to TS 38.101-1 on description of UE additional output power reduction, ZTE Corporation R4-1904929, draft Rel-15 CR for editorial corrections in 38.101-1, Ericsson R4-1904941, draft CR to 38.101-1 Correction to Pi/2 BPSK power boosting, Intel Corporation R4-1904957, Draft CR for TR38.101-1 – Update to EVM averaging, Rohde & Schwarz R4-1904958, Draft CR for TR38.101-1 – Update to spectrum flatness, Rohde & Schwarz R4-1904967, Draft CR for 38.101-1 definition of Maximum input level for intra-band contiguous CA, Huawei R4-1904969, Draft CR for 38.101-1: editorial correction, Huawei R4-1904987, Draft CR for correction on TS38.101-1, CATT Endorced draft CRs from RAN4#91 R4-1905339 removal of A-MPR brackets in FR1 Nokia R4-1905503 Change description 4.2(d) in Applicability of minimum requirements for TS 38.101-1 vivo R4-1905524 [Rx]Draft CR for 38.101-1 Removing the brackets in Rx requirements Huawei	
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					R4-1905526 [Rx]Draft CR for 38.101-1 defining NBB requirements<2.7GHz Huawei R4-1905772 Draft CR to TS38.101-1 Almost contiguous MPR Intel Corporation R4-1905795 Correction to a description of PRB for in-band emission in FR1 Anritsu Corporation R4-1905797 Correction to power control in FR1 Anritsu Corporation R4-1906140 draft CR for TS 38.101-1 Rx requirement for CA Huawei R4-1906153 Draft CR for TS 38.101-1: Editorial corrections to intra-band contiguous CA ACS and in-band blocking requirements MediaTek Inc. R4-1906154 Draft CR for TS 38.101-1: Adding symbol definitions for intra-band contiguous CA Rx maximum input level and ACS requirements MediaTek Inc. R4-1906871 Draft CR for TS 38.101-1 UE optional bandwidth for FR1 Huawei R4-1907131 Draft CR to 38.101-1. Clarification to FR1 NS_43 AMPR frequency ranges Qualcomm Incorporated R4-1907135 Draft CR to 38.101-1 rel. 15 to fix missing Exceptions for Out-of-band Blocking Apple R4-1907419 Draft CR for TS 38.101-1: Editorial improvement to EVM equalizer spectrum flatness requirements for Pi/2 BPSK MediaTek Inc. R4-1907429 Draft CR to TS38.101-1 A-MPR for Inter-band CA Intel Corporation R4-1907434 [Rx]Draft CR for 38.101-1 modifying characteristics of the interfering signal in Annex D Huawei R4-1907435 Draft CR to TS38.101-1 introduction of n41C and corrections on Rx requirements for NR intra-band contiguous CA ZTE Corporation R4-1907439 Draft CR to TS 38.101-1 on CA bandwidth class description ZTE Corporation R4-1907471 Draft CR to 38.101-1. Clarify all RB reference so transmission BW applies for all SCS Qualcomm Incorporated R4-1907474 Draft CR for TS 38.101-1 Correction of channel bandwidth set for NR CA Huawei R4-1907477 Draft CR to TS 38.101-1 on maximum aggregated bandwidth for NR CA configurations ZTE Corporation R4-1907481 Correction of RefSens exceptions due to UL harmonic interference for NR CA in 38.101-1 vivo R4-1907687 Correction to CA carrier spacing Ericsson	
2019-09	RAN#85	RP-192049	0078		F CR to TS 38.101-1: Implementation of endorsed draft CRs from RAN4#92 (Rel-15) R4-1907953 Correction to reference sensitivity for Band n74 Qualcomm Incorporated R4-1907955 Correction to reference sensitivity for Band n5 and n8 Qualcomm Incorporated	15.7.0

					R4-1907985 Update to FR1 EVM definition Rohde & Schwarz R4-1908022 Draft CR to 38.101-1 rel. 15 to fix Out-of-band Blocking issue for bands n51, n76 Apple R4-1908248 Draft CR to 38.101-1 NS_40, NS_41, NS_42 spurious emission requirement Intel Corporation R4-1908249 Draft CR to 38.101-1 A-MPR for NS_05 and NS_05U Intel Corporation R4-1908432 Further correction of RefSens exceptions due to UL harmonic interference for NR CA and SUL in 38.101-1 vivo R4-1908522 Draft CR for TS 38.101-1: Editorial corrections for transmit ON/OFF time mask MediaTek Inc. R4-1908523 Draft CR for TS 38.101-1: Corrections to inter-band CA and SUL OBB additional exception requirement MediaTek Inc. R4-1908572 Draft CR to TS 38.101-1: corrections on Rx requirements for intra-band CA ZTE Corporation, R4-1908632 Draft CR to TS38.101-1: Corrections on EVM window length (Section F.5) ZTE Corporation R4-1908707 Draft CR to TS38.101-1: corrections on the receiver spurious emission (section 7.9) ZTE Corporation R4-1908782 Draft CR: Correction to n70 UE Co-existence Dish Network R4-1908958 draft CR for 38.101-1: adding operating band for intra-band CA Huawei, HiSilicon R4-1908962 draft CR for 38.101-1: editorial correction for NBB, ACS and In-band emissions Huawei, HiSilicon R4-1909264 Draft CR to TS 38.101-1 on maximum output power reduction for PC3 ZTE corporation R4-1909737 Draft CR for editorial corrections in TS 38.101-1 Google Inc. R4-1909920 Draft CR for TS 38.101-1 Correction of referd table for NBB Huawei, HiSilicon R4-1909954 draftCR 38.101-1 Addition of footnote for n28 Skyworks Solutions Inc. R4-1909959 draftCR to 38.101-1 NS_05 AMPR RBstart correction for CIM3 Qualcomm Incorporated R4-1910247 Draft CR to add simultaneous RX/TX capability requirements in R15 TS 38.101-1 CMCC R4-1910253 draftCR to 38.101-1 NS_43 AMPR MPR threshold change and AMPR reduction Qualcomm Incorporated R4-1910289 Draft CR for 38.101-1: Correction to the Spurious Emission for UE Coexistence table Sprint Corporation, Ericsson, Google R4-1910326 Draft CR: REFSENS for SDL bands Qualcomm, Dish network R4-1910327 Draft CR to TS 38.101-1: Corrections for DL RMC for FR1 tests Intel Corporation R4-1910330 dCR to 38.101-1: Reference signal clarifications Qualcomm Incorporated R4-1910413 Draft CR for 38.101-1 correction for channel raster Huawei, HiSilicon R4-1910592 Draft CR for TS 38.101-1 SUL configured power	
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						correction InterDigital Communications R4-1910602 Draft CR for TS 38.101-1: Channel spacing for adjacent NR carriers Huawei R4-1910604 draft CR to TS38.101-1 Addition of a new note on UL_MIMO_for_SUL (section 5.2C) ZTE Wistron Telecom AB	
2019-12	RAN#86	RP-193028	0095		F	CR to 38.101-1: DMRS Exceptions	15.8.0
2019-12	RAN#86	RP-193028	0098		F	CR for 38.101- RX Out-of-Band Blocking for B38 and B41	15.8.0
2019-12	RAN#86	RP-193028	0102	1	F	CR for 38.101-1 n39 AMPR	15.8.0
2019-12	RAN#86	RP-193028	0111		F	CR on Sync raster to SSB resource element mapping	15.8.0
2019-12	RAN#86	RP-193028	0113		F	CR to TS 38.101-1 Almost contiguous A-MPR (R15)	15.8.0
2019-12	RAN#86	RP-193028	0117		F	CR to 38.101-1 (Rel-15) to clarify measurement interval and observation window on frequency error	15.8.0
2019-12	RAN#86	RP-193028	0120	1	F	CR to TS 38.101-1: Replace CBW with symbols defined in the specification	15.8.0
2019-12	RAN#86	RP-193028	0127	1	F	CR for TS38.101-1, Clarification and Editorial corrections	15.8.0
2019-12	RAN#86	RP-193028	0135		F	CR to TS 38.101-1 on A-MPR table cleanup (Rel-15)	15.8.0
2019-12	RAN#86	RP-193028	0136		F	CR for TS 38.101-1: Editorial correction for n2 uplink configuration note index in Table 7.3.2-3	15.8.0
2019-12	RAN#86	RP-193029	0139		F	CR for TS 38.101-1: Removing CA configurations for CA_n77D/E, CA_n78D/E, and CA_n79D/E	15.8.0
2019-12	RAN#86	RP-193029	0141	2	F	CR for TS 38.101-1: CA bandwidth class definition amendment	15.8.0
2019-12	RAN#86	RP-193029	0143		F	CR for TS 38.101-1: Fix out-of-band blocking issue for n50 and n75	15.8.0
2019-12	RAN#86	RP-193029	0145		F	CR to TS 38.101-1 on corrections to channel raster entries for NR band (Rel-15)	15.8.0
2019-12	RAN#86	RP-193029	0149	1	F	CR to transmit modulation quality in FR1	15.8.0
2019-12	RAN#86	RP-193029	0152		F	Removal of brackets from receiver requirements in 38.101-1 REL-15	15.8.0
2019-12	RAN#86	RP-193029	0156		F	CR to 38.101-1: Editorial correction of UL RMCs	15.8.0
2019-12	RAN#86	RP-193029	0158	1	F	CR for asynchronous operation for NR CA n78-n79	15.8.0
2019-12	RAN#86	RP-193029	0176		F	CR to 38.101-1-f70 Corrections to Transient Time Masks	15.8.0
2019-12	RAN#86					Table of contents updated	15.8.1
2019-12	RAN#86					Version number corrected on cover page	15.8.2
2020-03	RAN#87	RP-200394	0202		F	CR to TS 38.101-1 on corrections to network signalling value (Rel-15)	15.9.0
2020-03	RAN#87	RP-200394	0207		F	CR for 38.101- n39 NS flag change due to conflict	15.9.0
2020-03	RAN#87	RP-200394	0209		F	CR for 38.101-1: n41 and n25 corrections	15.9.0
2020-03	RAN#87	RP-200394	0217		F	CR to TS 38.101-1: corrections on ACS for intra-band contiguous CA	15.9.0
2020-03	RAN#87	RP-200394	0220			CR to TS 38.101-1: Replace CBW with symbols defined in the specification. NOTE: Corresponding Cat F CR to a non-implementable Rel-16 Cat A CR0221. The CR is not implemented in order to maintain consistency between Releases.	15.9.0

2020-03	RAN#87	RP-200394	0228	1	F	CR for TS38.101-1, Remove notes for UE channel bandwidth	15.9.0
2020-03	RAN#87	RP-200394	0230		F	CR for TS38.101-1, Correction of IE RF-Parameters name of maxUplinkDutyCycle	15.9.0
2020-03	RAN#87	RP-200394	0243		F	CR for 38.101-1: to remove fallback group 1 in table 5.5A.1-1	15.9.0
2020-03	RAN#87	RP-200394	0264		F	CR for [agreed] asynchronous operation for NR CA n78-n79	15.9.0
2020-03	RAN#87	RP-200394				CR for [agreed] asynchronous operation for NR CA n78-n79 NOTE: Corresponding Cat F CR to a non-implementable Rel-16 Cat A CR0275.The CR is not implemented in order to maintain consistency between Releases.	15.9.0
2020-03	RAN#87	RP-200394	0277		D	Removal of unnecessary definition of offsetmax,IMD3 from Table 6.2.3.2-1	15.9.0
2020-06	RAN#88	RP-200985	0299		F	CR to asymmetric CBW operation in FR1	15.10.0
2020-06	RAN#88	RP-200985	0326		F	Maintenance CR to 38101-1 on relative power tolerance R15	15.10.0
2020-06	RAN#88	RP-200985	0330		F	Update of CSI-RS definition for FR1 DL RMCs	15.10.0
2020-06	RAN#88	RP-200985	0334		F	Correction to FR1 QPSK UL RMC	15.10.0
2020-06	RAN#88	RP-200985	0339		F	CR to TS 38.101-1: Replace CBW with symbols defined in the specification.	15.10.0
2020-06	RAN#88	RP-200985	0353		F	IBE measurements for half Pi BPSK with spectrum shaping	15.10.0
2020-06	RAN#88	RP-200985	0375		F	CR for 38.101-1 to remove the NR CA configuration for REFSSENS exception due to cross band isolation for CA	15.10.0
2020-06	RAN#88	RP-200985	0378		F	IBE requirement for almost contiguous allocations	15.10.0
2020-06	RAN#88	RP-200985	0399		F	TS38.101-1 CR on 30KHz SSB SCS for n40	15.10.0
2020-06	RAN#88	RP-200985	0403		F	CR for 38.101-1: to add some missing sub-clause title for NR inter-band CA	15.10.0
2020-06	RAN#88	RP-200985	0337	1	F	CR to TS 38.101-1: Correction on the CA nominal channel spacing	15.10.0
2020-06	RAN#88	RP-200985	0342	1	F	CR for [agreed] asynchronous operation for NR CA n78-n79	15.10.0
2020-06	RAN#88	RP-200985	0376	1	F	CR for 38.101-1 to add the REFSSENS exception for inter band CA with SDL	15.10.0
2020-06	RAN#88	RP-200985	0384	1	F	OOB blocking for n70 adjacent to n25	15.10.0
2020-06	RAN#88	RP-200985	0297	1	F	Corrections of UE co-ex tables for Japan-related bands (R15)	15.10.0
2020-06	RAN#88	RP-200985	0309	1	F	CR to TS 38.101-1 R15: corrections on ACS for intra-band contiguous CA	15.10.0
2020-06	RAN#88	RP-200985	0301	1	F	CR on ACLR MBW definition in FR1	15.10.0
2020-06	RAN#88	RP-200985	0344	2	F	30k SSB SCS for n50	15.10.0
2020-06	RAN#88	RP-200985	0346	2	F	Addition of 30k SSB SCS for Band n38	15.10.0
2020-06	RAN#88	RP-200985	0350	2	F	Introduction of the Annex modifiedMPR-Behaviour into the NR SA specification	15.10.0
2020-09	RAN#89	RP-201512	0410		F	OOB blocking for Inter-band CA	15.11.0

2020-09	RAN#89	RP-201512	0425	1	F	CR to TS 38.101-1: corrections on narrow band blocking for intra-band contiguous CA	15.11.0
2020-09	RAN#89	RP-201512	0434		F	Corrections of Japan-related CA co-ex tables for REL-15 combo	15.11.0
2020-09	RAN#89	RP-201512	0441	1	F	30k SSB for n34 and n39	15.11.0
2020-09	RAN#89	RP-201512	0443		F	Correction for 5 MHz channel bandwidth for n40 and n50 (15k SCS)	15.11.0
2020-09	RAN#89	RP-201512	0457	1	F	CR for 38.101-1 RFC corrections (R15)	15.11.0
2020-09	RAN#89	RP-201512	0461	1	F	CR for 38.101-1 to add the missing MSD for CA_n41A-n78A	15.11.0
2020-09	RAN#89	RP-201512	0464		F	Correction to configured power with allowance for SRS switching	15.11.0
2020-09	RAN#89	RP-201512	0482		F	Correction of applicability of 2Rx requirements	15.11.0
2020-09	RAN#89	RP-201512	0490		C	7.5 kHz UL shift for LTE/NR spectrum sharing in Band 38/n38	15.11.0
2020-12	RAN#90	RP-202485	0494	1	F	CR to 38.101-1: UL MIMO EVM and emission requirements update	15.12.0
2020-12	RAN#90	RP-202485	0511	1	F	CR to TS38.101-1 on DC location correction	15.12.0
2020-12	RAN#90	RP-202485	0517		F	Coexistence cleanup for 38.101-1 Rel15	15.12.0
2020-12	RAN#90	RP-202485	0526	1	F	CR to TS 38.101-1[R15]: Clarification of non-simultaneous Rx/Tx operation for CA_n77-n79 and CA_n78-n79 in TS 38.101-1.	15.12.0
2020-12	RAN#90	RP-202485	0541		F	CR for 38.101-1 to adjust the structure of NR CA REFSSENS	15.12.0
2020-12	RAN#90	RP-202485	0555		F	CR Removal of Band 10 protection 38101-1 Rel15	15.12.0
2020-12	RAN#90	RP-202485	0570		F	CR for TS 38.101-1: correction of delta Tib for UE supporting multiple band combinations (R15)	15.12.0
2020-12	RAN#90	RP-202485	0580	1	F	CR on correction for AMPR NS_38,NS_40 and NS_41	15.12.0
2020-12	RAN#90	RP-202485	0582	1	F	CR to DMRS position in UL RMC for FR1	15.12.0
2021-03	RAN#91	RP-210117	0610		F	CR for TS38 101-1 Rel-15 Correction for definition of P-MPR	15.13.0
2021-03	RAN#91	RP-210117	0661		F	Simplification of n70	15.13.0
2021-03	RAN#91	RP-210117	0672		F	CR to TS38.101-1: Correction on applicability of minimum requirements	15.13.0
2021-03	RAN#91	RP-210117	0675	1	F	CR to TS38.101-1: Correction on the Aggregated Channel Bandwidth	15.13.0
2021-03	RAN#91	RP-210117	0096		F	CR for TS 38.101-1: Cleanup for spurious emissions for UE co-existence table	15.13.0
2021-03	RAN#91	RP-210117	0697	2	F	CR for TS 38.101-1: Correction to FR1 time mask for SRS antenna switching	15.13.0
2021-03	RAN#91	RP-210117	0718		F	Corrections to PCMAX for UL CA	15.13.0
2021-06	RAN#92	RP-211084	0734		F	Update of FR1 UL RMC tables	15.14.0
2021-06	RAN#92	RP-211085	0766		F	CR to TS38.101-1[R15]: Addition of UE co-existence requirements for n40	15.14.0
2021-06	RAN#92	RP-211080	0777	1	F	Cleanup for UE co-existence 38.101-1 Rel-15	15.14.0
2021-09	RAN#93	RP-211910	0909		B	Introduction of the UL 7.5kHz shift for NR TDD band n34 and n39	15.15.0
2021-09	RAN#93	RP-211921	0919		F	Big CR for TS 38.101-1 Maintenance part1 (Rel-15)	15.15.0
2021-12	RAN#94	RP-212856	0981		F	Big CR for TS 38.101-1 Maintenance (Rel-15)	15.16.0
2022-03	RAN#95	RP-220337	1035		F	Big CR for TS 38.101-1 Maintenance Part-1 (Rel-15)	15.17.0
2022-06	RAN#96	RP-221655	1119		F	Big CR for TS 38.101-1 Maintenance Part-1 (Rel-15)	15.18.0
2022-09	RAN#97	RP-222026	1190		F	Big CR for 38.101-1 maintenance part1 (Rel-15)	15.19.0

2022-12	RAN#98-e	RP-223290	1208		F	Addition of FR1 UL MIMO EVM measurement description	15.20.0
2022-12	RAN#98-e	RP-223290	1211		F	Addition of FR2 UL MIMO EVM measurement description Note: The CR was not implementable and therefore was not implemented in the specification.	15.20.0
2022-12	RAN#98-e	RP-223291	1265		F	CR on 'Annex G Difference of relative phase and power errors' for FR1 UL coherent MIMO	15.20.0
2022-12	RAN#98-e	RP-223291	1268	1	F	CR on TDD RMC for Intra-band EN-DC - TS 38.101-1	15.20.0
2023-03	RAN#99	RP-230504	1321	1	F	Addition of configuration for carrier aggregation RMCs	15.21.0
2023-03	RAN#99	RP-230504	1356	1	F	CR to clarify duplex mode of SDL bands	15.21.0
2023-03	RAN#99	RP-230502	1363		F	CR on Harmonic mixing MSD for CA_n8A-n79A (R15)	15.21.0
2023-03	RAN#99	RP-230504	1408	1	F	CR for TS 38.101-1 to clarify the inner outer condition for almost contiguous RB allocation	15.21.0
2023-03	RAN#99	RP-230503	1433		F	CR to TS 38.101-1 on humidity condition for normal temperature	15.21.0
2023-03	RAN#99	RP-230504	1456		F	Output power for NS_38, NS_40, and NS_41	15.21.0
2023-03	RAN#99	RP-230504	1468		F	CR to TS 38.101-1 Rel-15 Minimum guardband	15.21.0
2023-06	RAN#100	RP-231355	1481		F	CR to K1 and PdschNumOfHarqProcess for DL-CA	15.22.0
2023-06	RAN#100	RP-231355	1485		F	FR1 OOB requirements correction	15.22.0
2023-06	RAN#100	RP-231355	1540		F	CR for TS 38.101-1 on corrections to the minimum guardband calculation	15.22.0
2023-06	RAN#100	RP-231356	1602		F	CR to 38.101-1 Rel-15 Cat F, FRC correction	15.22.0
2023-06	RAN#100	RP-231356	1631	1	F	CR on correcting n38 UL requirement note 22	15.22.0
2023-09	RAN#101	RP-232487	1656		F	CR for TS 38.101-1 Rel-15 CAT-F: Introducing editorial modification for NS_43 A-MPR region	15.23.0
2023-09	RAN#101	RP-232505	1663	1	F	CR to clarify pi2BPSK note	15.23.0
2023-09	RAN#101	RP-232487	1689		F	CR for 38101-1: Almost contiguous NBC change reversal	15.23.0
2023-09	RAN#101	RP-232496	1726		D	[NR_RF_TxD] Editorial modification CR for TS 38.101-1 NOTE: This was not implemented because the CR coversheet state it is Rel-17 and not Rel-15, but CR was allocated for Rel-15	15.23.0
2023-09	RAN#101	RP-232487	1728		F	CR for Rel-15 38.101-1 to correct the superscript of inter-band CA_n77-n79	15.23.0
2023-09	RAN#101	RP-232502	1732		D	[NR_newRAT-Core] Editorial modification CR for TS 38.101-1_v2	15.23.0
2023-09	RAN#101	RP-232503	1747	1	F	[NR_newRAT-Perf] CR: Correction of FRC for maximum input level for 256QAM	15.23.0
2023-09	RAN#101	RP-232501	1755	1	F	[NR_newRAT-Core] CR for TS 38.101-1 to modify MSD due to harmonic mixing interference (R15)	15.23.0
2023-09	RAN#101	RP-232502	1765	1	F	[NR_newRAT-Core] Editorial correction to 6.2A.4 (Rel-15)	15.23.0
2023-09	RAN#101	RP-232502	1769		F	[NR_newRAT-Core] Update of FR1 UL MIMO EVM measurement description	15.23.0
2023-09	RAN#101	RP-232505	1783		F	CR to TS 38.101-1: correction of Pmax tolerance for 2Tx (Rel-15)	15.23.0
2023-09	RAN#101	RP-232501	1786		F	[NR_newRAT-Core] Correction of intraband contiguous CA ACS requirements	15.23.0
2023-09	RAN#101	RP-232501	1794		F	[NR_newRAT-Core] Correction to EVM measurement point	15.23.0
2023-12	RAN#102	RP-233332	1812	1	F	Fc terminology update	15.24.0

2023-12	RAN#102	RP-233335	1827		F	UL MIMO Spurious emissions per UE	15.24.0
2023-12	RAN#102	RP-233332	1852	3	F	[NR_newRAT-Core] CR for TS 38.101-1 Rel-15: Introducing missing MSD for harmonic mixing	15.24.0
2023-12	RAN#102	RP-233332	1872	1	F	[NR_newRAT] CR for clarification on applicability of Rx antenna number for Rx requirements for TS 38.101-1	15.24.0
2023-12	RAN#102	RP-233332	1900		F	[NR_newRAT-Core] CR to remove the word capable in power class 3 capable UE - TS38.101-1, Rel-15, Cat-F	15.24.0
2023-12	RAN#102	RP-233332	1926	1	F	[NR_newRAT-Core] CR for 38.101-1 to clarify the applicable bands for additional UTRA ACLR requirements. (R15)	15.24.0
2023-12	RAN#102	RP-233331	1956		F	[NR_newRAT] CR to 38.101-1 on FRC correction	15.24.0
2023-12	RAN#102	RP-233331	1960		F	[NR_newRAT] CR to 38.101-1 on FRC deletion for 5MHz 30 KHz	15.24.0
2023-12	RAN#102	RP-233340	1969		F	Correction of ?T_RxSRS for SRS resource set consisting of two SRS ports	15.24.0
2024-03	RAN#103	RP-240563	2079		F	(NR_newRAT-Core) Correct on the NOTE for harmonic mixing MSD valid test point	15.25.0
2024-03	RAN#103	RP-240563	2103		F	(NR_newRAT-Core) CR to correct "Supplement" to "Supplementary" in the definition of the suffix used for SUL - TS38.101-1, Rel-15, Cat-F	15.25.0
2024-03	RAN#103	RP-240564	2116		F	(NR_newRAT-Core) CR for TS 38.101-1 to correct the Finterferer (offset) for intra-band CA ACS and IBB requirements (R15)	15.25.0
2024-03	RAN#103	RP-240564	2147		F	(NR_newRAT-Core) CR to remove the word capable in power class related requirements	15.25.0
2024-03	RAN#103	RP-240564	2151		F	(NR_newRAT-Core) CR to TS38.101-1: Correction of band number in uplink configuration for reference sensitivity table	15.25.0
2024-06	RAN#104	RP-241382	2284		F	(NR_newRAT-Core) Correct the Pcmx tolerance for inter-band CA	15.26.0
2024-06	RAN#104	RP-241382	2345		F	(NR_newRAT-Core) CR for TS 38.101-1 on UE additional spurious emissions (R15)	15.26.0
2024-09	RAN#105	RP-242174	2388	1	F	CR for TS 38.101-1 Rel-15 correction on the terminology of emission bandwidth for NS_04	15.27.0
2024-09	RAN#105	RP-242174	2433		D	(NR_newRAT-core) CR for TS 38.101-1 R15 correction on AMPR for NS_10	15.27.0
2024-09	RAN#105	RP-242174	2448		F	(NR_newRAT-Core) CR to correct the definition of the symbol Nrb_agg and two symbols on same line - TS38.101-1	15.27.0
2024-09	RAN#105	RP-242191	2467		F	Cat F CR to TS 38.101-1 Rel-15 REFSSENS Corrections	15.27.0
2024-09	RAN#105	RP-242191	2473		F	Cat F CR to TS 38.101-1 Rel-15 Power Class 4 clean-up	15.27.0
2024-09	RAN#105	RP-242191	2485		F	(TEI) On missing BCS set definition for asymmetric FDD	15.27.0
2024-09	RAN#105	RP-242191	2486		F	(TEI) On missing BCS set definition for asymmetric TDD	15.27.0
2024-09	RAN#105	RP-242191	2491		F	Rel-15 SUL configuration correction for REFSSENS alignment with subsequent releases	15.27.0
2024-09	RAN#105	RP-242173	2500	1	F	CR to 38.101-1 Rel-15: Corrections of NR operating bands clause in FR1	15.27.0
2024-12	RAN#106	RP-243054	2545	1	F	(NR_newRAT-Core) CR for TS 38.101-1 reduction of UL CA UetoUe co-ex R15	15.28.0
2024-12	RAN#106	RP-243054	2577		F	(NR_newRAT-Core) CR to modify MBW definition - TS38.101-1	15.28.0

2025-09	RAN#109	RP-252378	2861	1	F	(NR_newRAT-Core) CR to TR 38.101-1 on the channel bandwidth list in the normative Annex D	15.29.0
2025-09	RAN#109	RP-252378	2995	1	F	CR on clarification of maximum number of RB's for n50 asymmetric UL/DL BW's	15.29.0